Rocket Engine Mount Strut for A Reusable Launch Vehicle

Mass Efficient Structural Design, Analysis and Test Planning

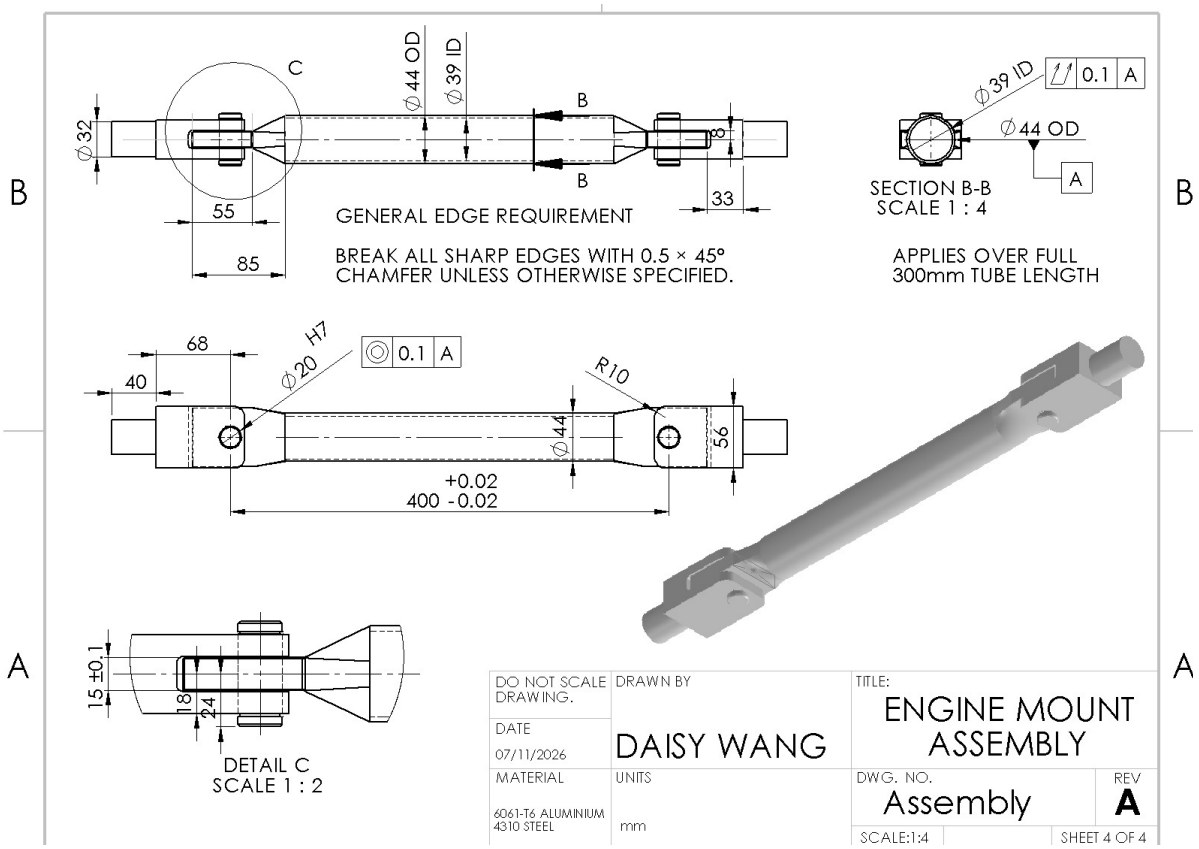
Daisy Wang | Personal Project | Sep 2026

Abstract

Designed a 0.43 scale, pin-ended 6061-T6 aluminium strut for a representative rocket engine mount, carrying 42.5 kN limit load in a 400 mm test article. The project covers the design process from requirement definition and structural sizing through CAD, analytical verification, FEA and physical test.

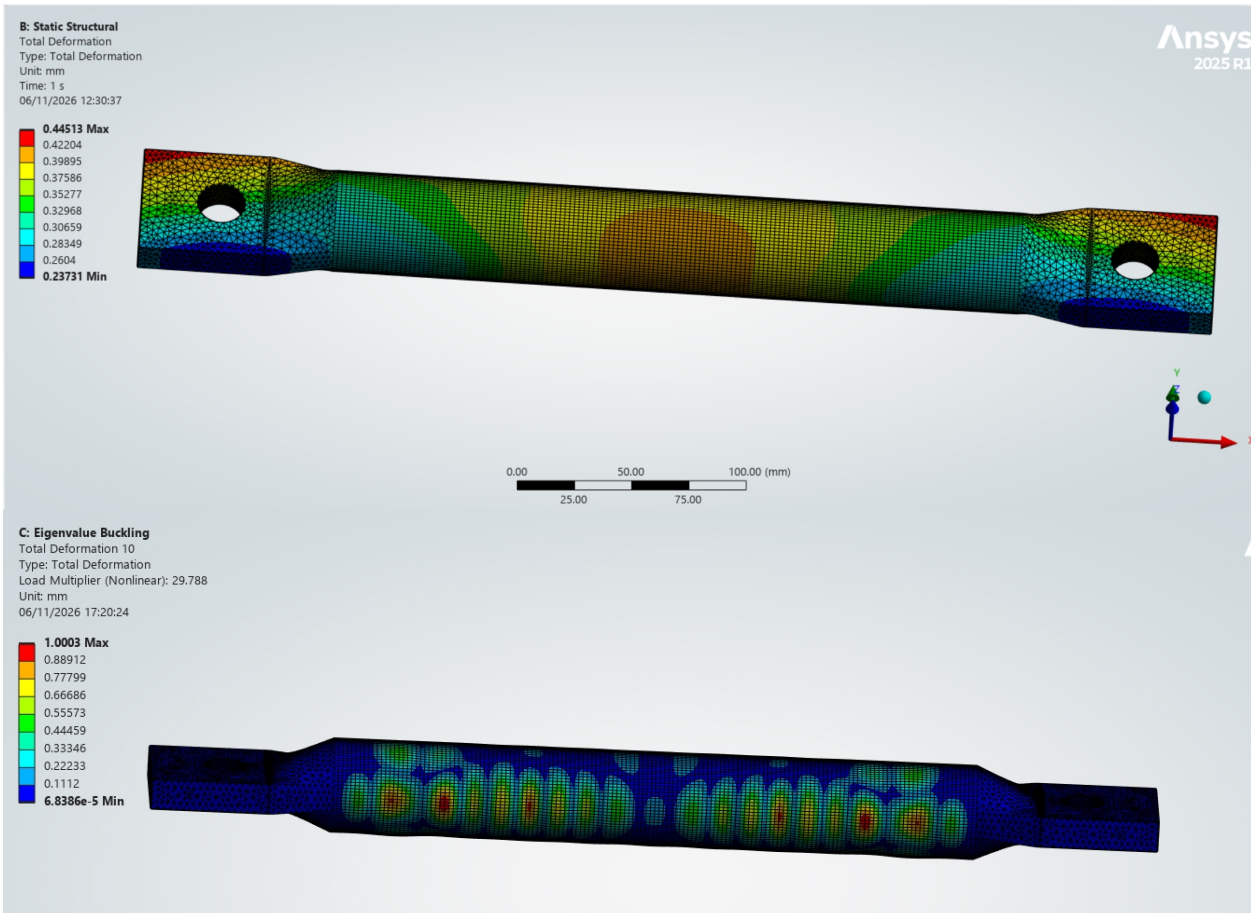
A parametric Python trade study evaluated tube diameter and wall thickness against yielding, ultimate strength, global buckling and local wall slenderness. The selected 44 mm × 2.5 mm tube is 36% lighter than the lightest solid round bar satisfying the same design criteria. The end fittings and pin were sized separately to ensure the strut remains the intended failure location.

The resulting strut has a mass of 0.805 kg and a predicted failure load of 70–83 kN, depending on load eccentricity. ANSYS results were cross checked against hand calculations, with differences of 2.4% in midspan stress and deflection. Mesh refinement changed the monitored results by no more than 0.2%. The analysis predicts yielding before elastic buckling. An instrumented test to failure is planned to correlate the predictions with measured behaviour and inform subsequent redesigns.



Key Engineering Results

- **Load definition:** Derived 42.5 kN limit and 59.5 kN ultimate loads for the 0.43 scale test article from engine thrust, mount geometry and load sharing. Full scale load: 224.8 kN per strut.
- **Mass optimisation:** Selected a 44 mm OD × 2.5 mm wall tube through a Python trade study. At 0.352 kg, it is 36% lighter than the equivalent solid bar.
- **Structural margins:** Achieved +0.30 yield, +0.40 ultimate and +0.12 buckling margins of safety. Local wall buckling stress is 21 times design yield strength.
- **Controlled failure sequence:** Sized end fittings to withstand 1.25 times predicted strut failure load and pins to remain elastic to 1.10 times that load.
- **Analysis verification:** ANSYS agreed with hand calculations within 2.4% for midspan stress and deflection. Mesh refinement changed monitored results by no more than 0.2%.
- **Failure assessment:** Predicted elastic buckling at 253 kN, approximately three times the peak failure load, making yielding the expected failure mode. Assessed the mesh sensitive bore stress peak through a corner radius study.
- **Test readiness:** Defined fabrication, inspection, instrumentation and test procedures to support a controlled test to failure and subsequent model correlation.

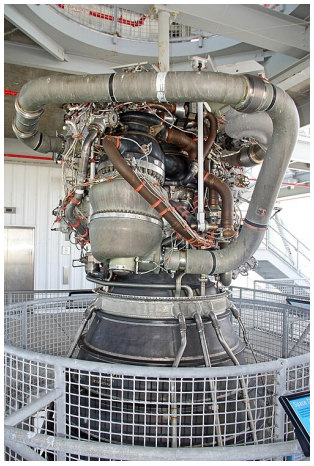


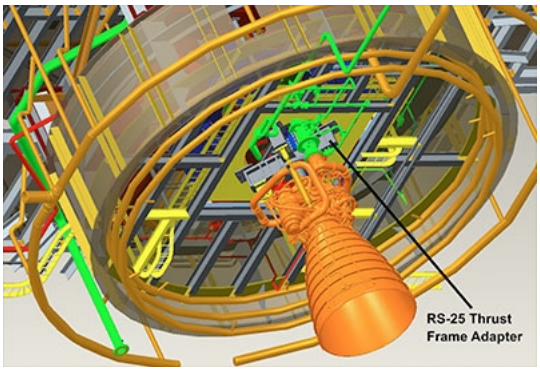
1. Introduction

1.1 Background

A launch-vehicle engine mount must carry thrust, resist engine gimbal and inertial loads, and weigh as little as possible. Across the industry, vehicles connect engines to structure in a few recurring ways:

1. The gimbal-bearing pedestal (SLS/RS-25, Shuttle). Each RS-25 attaches to the vehicle through a titanium gimbal bearing — a universal ball-and-socket joint, bolted to the launch vehicle by its upper flange and to the engine by its lower flange, supporting 3,390 kg of engine weight and over 500,000 lbf of thrust while pivoting ±10.5°.





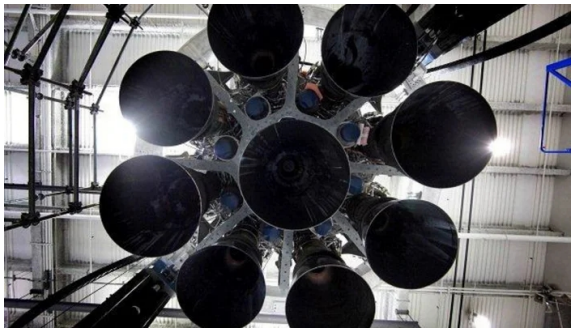
2. The strut/fitting truss ("thrust cone") — Saturn V F-1, and the internal structure inside modern thrust sections. Engines hang from a conical or ring frame through pin-jointed struts and machined end fittings. This is why the F-1 cluster photos show a conical lattice above the nozzles, and SLS CAD shows an "RS-25 Thrust Frame Adapter."

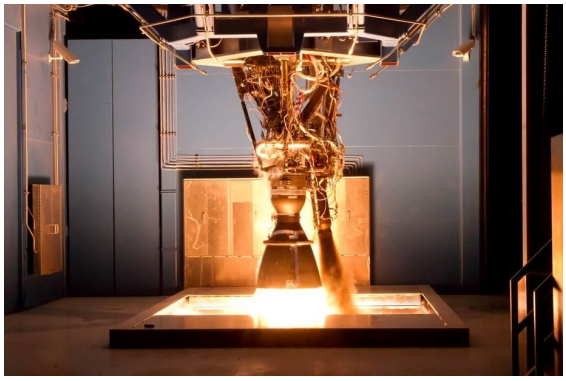




3. The ring-frame cluster (SpaceX Octaweb). Eight Merlins in a circle around a center engine on a machined aluminum structure that "reduces the length and weight of the Falcon 9 thrust structure and

streamlines manufacturing." Look at the base of the booster: the engines bolt into machined pockets of one monolithic aluminum frame.





4. Hold-down/release fittings. At the pad, the vehicle is restrained against release mechanisms (visible as the arm-like fittings gripping the Falcon 9 base below) — a deliberately fusible, high-load joint that is its own whole discipline.



The common thread: every one of these is built from **two-force members (struts) + pinned lug/clevis fittings + a frame**. A strut is a good test article because every failure mode has a closed-form prediction, and it can be broken in an ordinary compression test.

1.2 Research question and objectives

Research question. How can a lightweight engine support attachment meet a defined strength and stiffness requirement while minimizing mass, and how accurately can hand calculations and FEA predict its actual failure?

Objectives:

1. Define a load case and requirements traceable to published sources.
2. Size a minimum-mass strut that meets the strength and stability criteria, using a two-variable sweep.
3. Design end fittings so that the strut column, not the joint, is the first element to fail.
4. Predict the stiffness, the failure load and the failure mode by hand calculation and FEA.
5. Test the part to failure, quantify the prediction error, and redesign from the observed failure mechanism.

1.3 Summary of results

The selected strut is a tube of 44 mm outside diameter and 2.5 mm wall, 300 mm long, with 56 mm diameter solid ends tapering into 15 mm thick tongues drilled for 20 mm pins. Pin-to-pin length is 400 mm and the mass is 0.805 kg. At the design limit load of 42.5 kN the secant peak stress is 146 MPa (e = 1 mm). Margins at 1.25 and 1.4 times limit load are +0.30 and +0.40, and the buckling margin is +0.12 against a 10% requirement. The predicted peak load is 70 to 83 kN, and the ends are sized to carry at least 1.25 times the strut's predicted maximum. A finite element model of the worst eccentricity (2 mm) gives mid-length stresses of 160 and 96 MPa against 162 and 98 MPa by hand, an added deflection of 0.400 mm against 0.410 mm, and a first elastic buckling load of 253 kN, three times the predicted peak load.

2. Strut concept and configuration

2.1 Definitions

The **strut** (the test specimen) is a single machined part with a hollow tubular shank, two solid tapered transitions and two flat **tongues** at its ends. Each tongue has a reamed hole for a pin. The **fixture** is a steel clevis at each end: two fork plates and a **pin** in double shear. The tongue is the inner lug and the fork plates are the outer lugs of the double-shear joint. The pin axis passes through the tongue thickness. Rotation about the pin axis is free, so the shank is pinned in the plane perpendicular to the pin axis. The forks restrain rotation about the other axes, and the analysis conservatively treats both planes as pinned.

This arrangement keeps the pin joint out of the column section. The tongue is wider than the tube wall can carry as a column, the hole sits in a short solid region that cannot buckle as a column, and the shank is sized purely by column behaviour. There is no second joint between the tongue and the shank, because they are one part.

2.2 Load path and free-body diagrams

Figure 1.(a) shows the load path. Engine thrust T is shared by N inclined struts. Each strut is a two-force member: the force is equal and opposite at its two pins and acts along the line joining them. In the ideal case the strut carries a pure axial load. In practice the load line is offset from the centroid by an eccentricity e, and the strut bends as it is loaded, which is analysed in a section on buckling. At the pin joint, the strut tongue transmits P to the pin and the pin transmits P/2 to each fork plate. The pin bending arm in this joint governs the pin design.

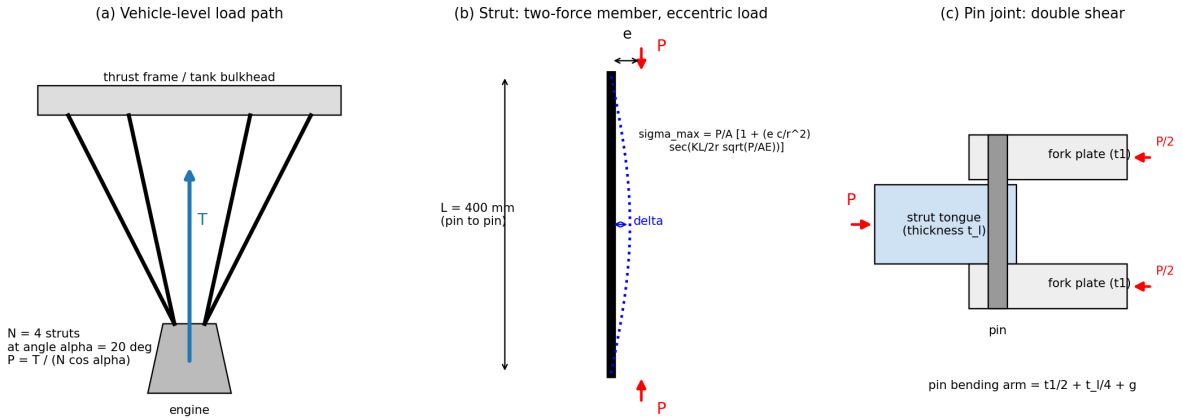


Figure 1.(b) shows the strut geometry.

2.3 Failure-order design rules

Since the test is intended to answer the research question, the failure sequence is controlled. The strut column should fail first, and the end hardware should remain elastic. Four design rules enforce this:

1. The shank meets all strength and stability criteria of Section 3.3 at design-minimum properties.
2. The tongues, holes and pins carry at least 1.25 times the strut's predicted maximum capacity, the Johnson column load at typical properties.
3. The pins stay elastic up to 1.10 times that maximum, so pin bending does not alter the measured load eccentricity.
4. The fixture and test machine are rated for at least 1.5 times that maximum.

3. Requirements and design basis

3.1 Material and pin data

Quantity	Value	Source
6061-T6 elastic modulus E	68.9 GPa	ARRK datasheet [8]
Shear modulus G	26 GPa	[8]
Density	2.70 g/cm3	[8]
Yield / ultimate strength, typical	276 / 310 MPa	[8], MatWeb extract [9]
Yield / ultimate strength, design minimum	240 / 290 MPa	6061-T6 minimums [10]
Shear strength	207 MPa	[8]

Quantity	Value	Source
Bearing ultimate / yield (edge distance / diameter = 2)	607 / 386 MPa	[9]
Pin steel (4130): Ftu, Fty, Fsu	861.8, 710.2, 565.4 MPa	Values printed in the worked example of the Air Force Stress Manual [14]
Thrust and diameter, Merlin 1D (sea level)	845 kN, 0.92 m	Wikipedia [1]
Ultimate factor of safety	1.4	NASA-STD-5001B [7]
Yield factor of safety	1.25	[7] (protoflight value)

Design-minimum strengths are used for all pass/fail criteria. Typical strengths are used only to predict the most probable test outcome.

3.2 Requirements

ID	Requirement
R1	Maximum secant stress at 1.25 x limit load does not exceed the design yield strength (240 MPa)
R2	Maximum secant stress at 1.4 x limit load does not exceed the design ultimate strength (290 MPa)
R3	The column critical load at design-minimum properties is at least 1.10 x the ultimate load (buckling margin of at least 10%, consistent with published launch-structure practice [23])
R4	Tube wall: D/t at most 20, t at least 2.0 mm, and classical local buckling stress at least 3 x design yield
R5	Tongues, holes and pins carry at least 1.25 x the strut's predicted maximum; pins elastic to 1.10 x
R6	Axial stiffness is measured and reported
R7	Predicted peak load is within 15% of the measured value when the prediction uses measured eccentricity and material strength

3.3 Modelling assumptions

Assumption	Value	Basis
Number of struts N	4	Representative of a four-strut mount
Angle α from the thrust axis	20 degrees	Modeling assumption

Assumption	Value	Basis
Full-scale strut length	Equal to engine diameter (920 mm)	Sets the geometric scale factor
Test strut length L (pin centre to pin centre)	400 mm	Bench-test scale
End factor K	1.0	Pinned ends, conservative for the plane the forks restrain
Load eccentricity e	1.0 mm	Tolerance stack (Section 10.3); sensitivity 0.5 to 5 mm
Maximum tube outside diameter	50 mm	Envelope assumption for the sweep
Wall slenderness and minimum wall	D/t at most 20, t at least 2.0 mm	Robustness to machining and handling
Local buckling criterion	Classical stress at least 3 x design yield	Allows a shell knockdown factor down to one third; see Section 5.4
Fork thickness and pin gap	10 mm each and 0.5 mm per side	Fixture design
Lug coefficients K_n and K	0.74 and 1.33	Lowest values in the Air Force manual's worked examples; to be replaced from its figures

4. Load definition

Define a dimensionless geometric scale factor, λ. The outer diameter of the engine (0.92m for the Merlin 1D) is publicly known, standardized dimension, it is taken to be the characteristic dimension. λ is defined as the ratio of test strut length L to engine diameter D_{eng}:

$$\lambda = \frac{L}{D_{eng}} = \frac{400}{920} = 0.4348, \qquad \lambda^2 = 0.1890$$

Four struts inclined at 20 degrees to the thrust axis share the thrust T. Resolving force balance along the thrust axis:

$$P_{full} = \frac{T}{N \cos \alpha} = \frac{845{,}000}{4 \times 0.9397} = 224{,}800 \text{ N}$$

Geometric similarity keeps stress constant if force scales with area, i.e. with λ squared:

$$P_L = \lambda^2 P_{full} = 0.1890 \times 224{,}800 = 42{,}500 \text{ N}$$

The factored loads are the yield load P_y = 1.25 kN, P_L = 53.1 kN and the ultimate load P_u = 1.4kN, P_L = 59.5 kN. The load case is axial thrust only. Gimbal side load and engine inertia are outside the scope of this test and are treated as extensions (Section 12).

5. Analytical methods

5.1 Tube section properties

For a round tube of outside diameter D and wall t, with inside diameter d = D - 2t:

$$A = \frac{\pi}{4}\left(D^2 - d^2\right), \qquad I = \frac{\pi}{64}\left(D^4 - d^4\right), \qquad r = \sqrt{I/A}, \qquad c = \frac{D}{2}$$

The round section has the same radius of gyration in every plane, so there is a single buckling plane to check.

5.2 Global buckling

With slenderness S = K L / r, the Euler load and the transition slenderness are:

$$P_E = \frac{\pi^2 E I}{(KL)^2}, \qquad C_c = \sqrt{\frac{2\pi^2 E}{F_{ty}}}$$

Below C_c the Johnson parabola applies, and above it Euler applies:

$$\sigma_{cr} = F_{ty} - \frac{F_{ty}^2}{4\pi^2 E}\, S^2 \quad (S < C_c), \qquad \sigma_{cr} = \frac{\pi^2 E}{S^2} \quad (S \geq C_c), \qquad P_{cr} = \sigma_{cr} A$$

C_c is 75.3 at the design yield strength and 70.2 at the typical yield strength.

5.3 Eccentric loading

The load line is offset from the centroid by e. The secant formula gives the maximum stress and the additional lateral deflection at mid-length ([11]):

$$\varphi = \frac{KL}{2r}\sqrt{\frac{P}{AE}}, \qquad \sigma_{max} = \frac{P}{A}\left[1 + \frac{e\,c}{r^2}\sec\varphi\right], \qquad \delta_{max} = e\left(\sec\varphi - 1\right)$$

The formula is valid for pinned-pinned columns [12]. For a column that yields before it buckles, the useful failure indicator is the load at which σ_{max} reaches the yield strength. That load is always below the critical load.

5.4 Local buckling

The classical axial buckling stress of a thin cylindrical shell of mean radius R and wall t is 0.605 E t / R, reduced in practice by an imperfection knockdown factor γ (NASA SP-8007 method [16], formula summary [24]):

$$\sigma_{cl} = 0.605\,\gamma\,\frac{E\,t}{R}, \qquad \gamma = 1 \text{ for the classical value}$$

The criterion adopted here requires the classical value to exceed three times the design yield strength, which tolerates a knockdown factor down to one third. For the selected tube it exceeds the yield strength by a factor of 20.9, so local buckling does not govern.

5.5 Pin joint

The tongue-and-pin joint is analysed with the double-shear equations of the Air Force Stress Manual [14]. If e_h is too small, the high bearing load from the pin can shear or tear the material out right through the end of the tongue. This shear-out or net-section failure only happens in tension.

Here D is the hole and pin diameter, t_l the tongue thickness, e_h the distance from hole centre to tongue end, w the tongue width, t_1 the thickness of each fork plate, and g the gap per side

Bearing Capacity P_{brg} calculates how much crushing load the cylindrical wall of the hole can withstand before the pin permanently deforms or crushes the tongue material over $\mathrm{D} \times t_l$, the projected bearing area).

$$\mathrm{P}_{brg} = \min(K F_{tu},\, 1.304 K F_{ty},\, F_{bru},\, 1.304 F_{bry})\, D\, t_l$$

Net-Section Capacity P_{net}: when the pin pulls on the tongue, it creates tension across the remaining "meat" of the material on either side of the hole. The effective width is the total tongue width minus the hole diameter $(\mathrm{w} - \mathrm{D})$, multiplied by the tongue thickness t_l.

$$P_{net} = K_n F_{tu} (w - D)\, t_l$$

Pin Structural Capacities $\mathrm{P}_{ub,pin}, P_{el,pin}, P_{us,pin}$:

$\mathrm{P}_{ub,pin}$ (Ultimate Bending): The maximum bending moment the steel pin can handle before failing.

$\mathrm{P}_{el,pin}$ (Elastic Limit): The load limit where the pin stays completely elastic without taking a permanent bend, which is set to $1.10 \times$ the strut's max capacity.

$\mathrm{P}_{us,pin}$ (Double Shear): The load required to snap the 20 mm steel pin cleanly across its two shear planes ($1.571 D^2 F_{su,P}$).

$$b_{arm} = \frac{t_1}{2} + \frac{t_l}{4} + g$$

$$P_{ub,pin} = \frac{0.1963\, k_b\, D^3 F_{tu,P}}{b_{arm}}$$

$$P_{el,pin} = \frac{0.1963\, D^3 F_{ty,P}}{b_{arm}}$$

$$P_{us,pin} = 1.571\, D^2 F_{su,P}$$

The bearing and net-section equations apply when e_h / D is at least 1.5, and the plastic bending coefficient k_b is 1.56 for a ductile pin. The manual folds shear tear-out into the bearing coefficient K.

5.6 Compression loading of a tension-formulated lug

The manual's lug equations describe lugs loaded in tension. In compression the pin bears on the shank side of the hole, and the material between the hole and the free end is unloaded, so tear-out toward the

free end is not the expected mode. No source found for this work gives compression allowables. The tension equations are therefore used as a conservative bound, which also covers load reversal in service. The test applies compression only.

6. Shank sizing

6.1 Sizing Methodology and Design Space Definition

To identify the minimum-mass tubular shank, a two-variable parametric sweep was conducted over the outer diameter (D) and wall thickness (t).

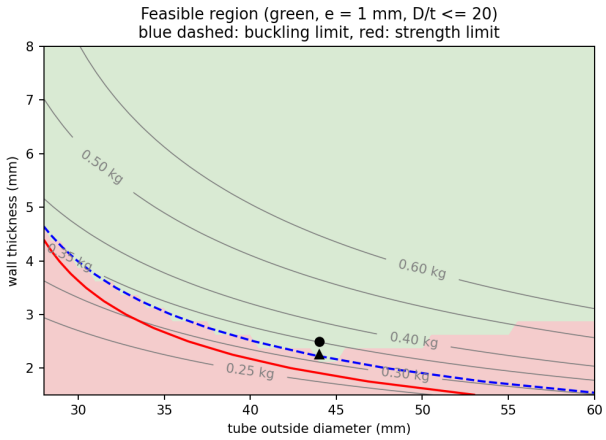
Design Space Boundaries:

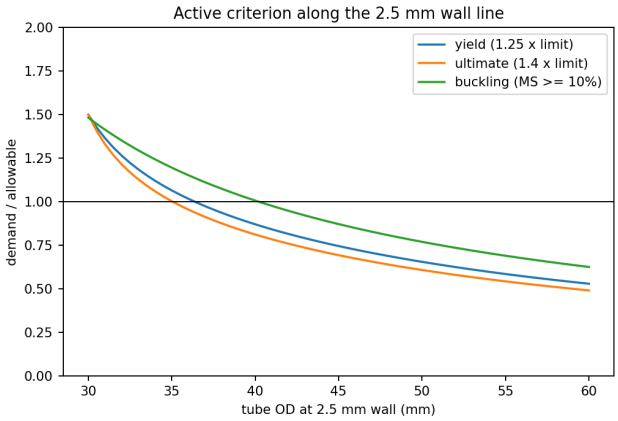
Outside Diameter $(28 \text{ to } 60 \text{ mm})$: The lower bound (28 mm) starts just above the lightest solid bar diameter (25.5 mm) to ensure it evaluates viable hollow alternatives. The upper bound (60 mm) is chosen to be comfortably larger than any reasonable optimum, ensuring the search space is fully bracketed.

Wall Thickness $(1.5 \text{ to } 12 \text{ mm})$: 1.5 mm is chosen as a lower bound approaching the minimum practical wall thickness for machining and handling without risking burn-through or excessive local shell buckling. 12 mm is an intentionally thick upper bound that approaches a solid bar.

Packaging and Manufacturing Constraints: To respect vehicle-level spatial clearance and prevent geometric interference within the engine bay, a strict upper envelope limit of $\mathrm{D} \leq 50 \text{ mm}$ was enforced. Additionally, a minimum wall thickness constraint $(\mathrm{t} \geq 2.0 \text{ mm})$ and a slenderness ceiling $(\mathrm{D}/\mathrm{t} \leq 20)$ were implemented to guarantee structural robustness against handling damage, tolerance stack-ups during deep-hole boring, and premature local shell buckling.

Evaluation Logic: For every valid coordinate pair (D, t), the model evaluates structural criteria R1 through R4 at the design-minimum material properties. Feasible configurations are filtered, and the global optimum is identified by selecting the geometry that minimizes cross-sectional area over the fixed 400 mm pin-to-pin length. Figure 3 shows the feasible region and the active criteria.





6.2 Results and sensitivity

With D/t at most 25 and t at least 2.0 mm:

e (mm)	Maximum OD (mm)	Optimum OD x wall (mm)	Area (mm2)	Mass over 400 mm (kg)	Governing criterion
0.5, 1.0 or 2.0	40	38.0 x 2.75	304.5	0.329	Buckling margin
0.5, 1.0 or 2.0	50 or 60	48.0 x 2.00	289.0	0.312	Buckling margin
5.0	40	39.5 x 3.50	395.8	0.428	Yield
5.0	50 or 60	50.0 x 2.25	337.5	0.365	Yield

With e = 1 mm and OD at most 50 mm, the optimum depends on the wall-slenderness bound:

D/t limit	Optimum OD x wall (mm)	Area (mm2)	Mass over 400 mm (kg)	Buckling utilization
15	38.0 x 2.75	305	0.329	0.98
20	44.0 x 2.25	295	0.319	0.99
25 or 30	48.0 x 2.00	289	0.312	1.00

Conclusions:

- Column Buckling Governs Design Limits: As illustrated in Figure 3, global column buckling (Constraint R3) forms the active constraint boundary for all realistic load eccentricities ($e \leq 2.0\ \mathrm{mm}$). Material yield only governs under extreme misalignments ($e \geq 5.0\ \mathrm{mm}$). Because buckling dominates within the expected tolerance range ($0.5\ \mathrm{to}\ 2.0\ \mathrm{mm}$), the theoretical optimum mass remains essentially invariant at approximately $0.32\ \mathrm{kg}$.

- Low Mass Sensitivity to Wall Slenderness $(\mathrm{D/t})$: evaluating the feasible region shows that relaxing the wall-slenderness constraint $(\mathrm{D/t})$ from 15 to 25 yields only a $5\%(17\ \mathrm{g})$ mass reduction (decreasing shank mass from $0.329\ \mathrm{kg}$ to $0.312\ \mathrm{kg}$). Because structural efficiency is insensitive to wall thickness within this regime, selecting a conservative ratio ($\mathrm{D/t} = 17.6, \mathrm{t} = 2.5\ \mathrm{mm}$) imposes a negligible mass penalty while significantly improving manufacturing robustness and deep-hole boring tolerance.
- Immunity to Local Shell Buckling: across all feasible sweep coordinates, the classical local shell-buckling utilization remains below 0.20, providing a margin of safety greater than 4.0. Localized wall crimping is therefore non-governing, confirming that wall thickness selection can be dictated entirely by handling durability and global column stability.

6.3 Selected shank

The optimum for D/t at most 20 is 44.0 x 2.25 mm. The wall is rounded up to 2.5 mm so that a bore tolerance of the order of 0.1 mm does not remove a significant fraction of the wall. The selected section is a tube of 44.0 mm outside diameter, 2.5 mm wall and 39.0 mm inside diameter:

$$A = 325.9\ \mathrm{mm}^2, \quad I = 70{,}423\ \mathrm{mm}^4, \quad r = 14.699\ \mathrm{mm}, \quad S = \frac{400}{14.699} = 27.21, \quad C_c = 75.28$$

Since S is below C_c, the Johnson branch applies. The Euler load (299 kN) corresponds to a stress of 918 MPa, which exceeds the yield strength and is not physical for this stocky column.

$$\sigma_{cr} = 240 - \frac{240^2}{4\pi^2(68{,}900)}(27.21)^2 = 224.3\ \mathrm{MPa}$$

$$P_{cr} = 224.3 \times 325.9 = 73.1\ \mathrm{kN}, \qquad MS = \frac{73.1}{1.1 \times 59.5} - 1 = +0.12$$

At the limit load (42.5 kN):

$$\varphi = \frac{400}{2(14.699)}\sqrt{\frac{42{,}500}{325.9 \times 68{,}900}} = 0.5919$$

$$\sec\varphi = 1.2050$$

$$\delta_{max} = 0.205\ \mathrm{mm}$$

$$\sigma_{max} = 130.4\left[1 + \frac{(1.0)(22)}{14.699^2}(1.2050)\right] = 146.4\ \mathrm{MPa}$$

Load case	P (kN)	P/A (MPa)	sec	Peak stress (MPa)	Allowable (MPa)	Margin
Limit	42.50	130.4	1.2050	146.4	n/a	n/a
1.25 x limit	53.12	163.0	1.2676	184.0	240	+0.30
1.4 x limit	59.50	182.5	1.3078	206.8	290	+0.40

The wall slenderness is D/t = 17.6, and the classical local-buckling stress is 5,022 MPa, 20.9 times the design yield strength.

6.4 Comparison with a solid bar

The lightest solid round bar that meets R1 to R3 has a diameter of 25.5 mm (511 mm^2 area, 0.552 kg over 400 mm). The selected tube has 0.352 kg over the same length, 36% lighter, and the sweep optimum (0.319 kg) is 42% lighter.

7. End-fitting design

7.1 Requirements

The strut's predicted maximum capacity is the Johnson load at typical properties, 83.2 kN. Rules 2 and 3 of Section 2.3 give:

$$P_{end} = 1.25 \times 83.2 = 104.0 \text{ kN}, \qquad P_{pin,elastic} = 1.10 \times 83.2 = 91.5 \text{ kN}$$

7.2 End-Fitting Selection

End fittings are sized to carry at least $1.25\times$ the strut's predicted maximum capacity (104.0 kN).

Python is again used to evaluate the end fittings using the following specific increments/ranges:

Head bar stock diameters (D_h): Discrete set of standard sizes: 50,56,60, and 63.5 mm.

Pin diameters (D_p): Discrete set of sizes: 16,18,20,22,25,28, and 30 mm.

Tongue thicknesses (t_l): Swept from 10 mm up to $0.8 \times D_h$ using an increment of 1.0 mm

The selected configuration, optimized for minimum volume, utilizes a 56 mm stock diameter, a 20 mm pin, and a 15 mm thick tongue.

Head Stock Selection and Net-Section Justification

The tongue is formed by milling two flats on the bar, so its width is $w = \sqrt{D_{head}^2 - t_l^2} = \sqrt{56^2 - 15^2} = 54.0$ mm.

The edge distance is 1.6 pin diameters rounded up to the next 5 mm. $1.6 \times D_{pin} = 1.6 \times 20 \text{ mm} = 32$ mm. Rounding up to the next 5 mm gives $e_h = 35$ mm $(e_h/D = 1.75)$

End fittings require a minimum capacity of $P_{end} = 104.0$ kN to ensure the column fails first. Machining tongues directly from 44.0 mm tube stock ($t_l = 15.0$ mm, $D_p = 20.0$ mm) yields a chord width of $w = \sqrt{44.0^2 - 15.0^2} = 41.36$ mm. This produces a net-section capacity of $P_{net} = K_n F_{tu}(w - D_p)t_l = 0.74 \times 290 \times (41.36 - 20) \times 15 = 68.9$ kN, which fails the 104.0 kN requirement. Heads are therefore turned from a larger 56.0 mm bar stock ($w = 54.0$ mm), increasing capacity to 109.3 kN.

Result summary table

Parameter	Value
Head bar stock diameter	56 mm
Pin and hole diameter	20 mm (H7 hole, g6 pin)
Tongue thickness	15 mm
Tongue width (chord)	54.0 mm
Edge distance e_h (hole centre to end)	35 mm (e_h / D = 1.75, a = 25.0 mm)
Tongue zone length	55 mm
Solid taper from 56 mm to 44 mm diameter	30 mm
Tube length	300 mm
Overall length	470 mm (pin centre distance 400 mm)
Fork plates and pin gap	10 mm each, 0.5 mm per side

7.3 Capacities and margins

The pin bending arm is 10/2 + 15/4 + 0.5 = 9.25 mm. Capacities with design-minimum properties (Section 3.1):

Mode	Capacity (kN)	Required (kN)	Margin vs requirement	Margin at ultimate design load (59.5 kN)
Tongue bearing	115.7 (123.7 typical)	104.0	+0.11	+0.94
Tongue net section	109.3 (116.8 typical)	104.0	+0.05	+0.84
Pin double shear	355	104.0	+2.42	+4.97
Pin ultimate bending	228	104.0	+1.19	+2.84
Pin elastic limit	121	91.5	+0.32	+1.28 (against the 53.1 kN yield load)

A two-plane tear-out estimate gives a high consistency margin of $170\ \mathrm{kN}$, though shear-out is inherently covered by the manual's bearing coefficient. Using typical material properties, tongue bearing and net-section capacities exceed the strut's maximum predicted load by 1.49 times and 1.40 times, respectively, while the pin elastic limit provides a 1.45 times margin against the yield load.

7.4 Fit

The pin and hole utilize an H7/g6 sliding fit. For the $18 \text{ to } 30 \text{ mm}$ size band, standard fit tables [19] specify a hole tolerance of $0 \text{ to } +21 \ \mu\text{m}$ and a shaft tolerance of $-20 \text{ to } -7 \ \mu\text{m}$, resulting in a controlled radial clearance of $7 \text{ to } 41 \ \mu\text{m}$. A second reference [20] confirms the H7 hole band and describes H7/g6 as the sliding fit. The pin is therefore a sliding fit, 19.980 to 19.993 mm in a hole of 20.000 to 20.021 mm. This mechanical fit ensures smooth assembly while contributing a negligible maximum of $20.5 \ \mu\text{m}$ to total load eccentricity.

7.5 Mass and stiffness

The total part mass is 0.805 kg, broken down into 0.264 kg for the tube, 0.320 kg for the solid tapers, and 0.221 kg for the end tongues. Consequently, the heavy end regions account for 67% of the total mass. A segmented-bar model predicts an axial stiffness of 61.9 kN/mm (56.1 kN/mm for the bare tube over the 400 mm span). Because this model excludes pin, hole, and test machine compliance, actual measured laboratory stiffness is expected to be slightly lower.

8. Predicted Behaviour

First yield is the load at which the secant peak stress reaches the typical yield strength of 276 MPa. When the strut reaches its Johnson load before first yield, the Johnson load governs.

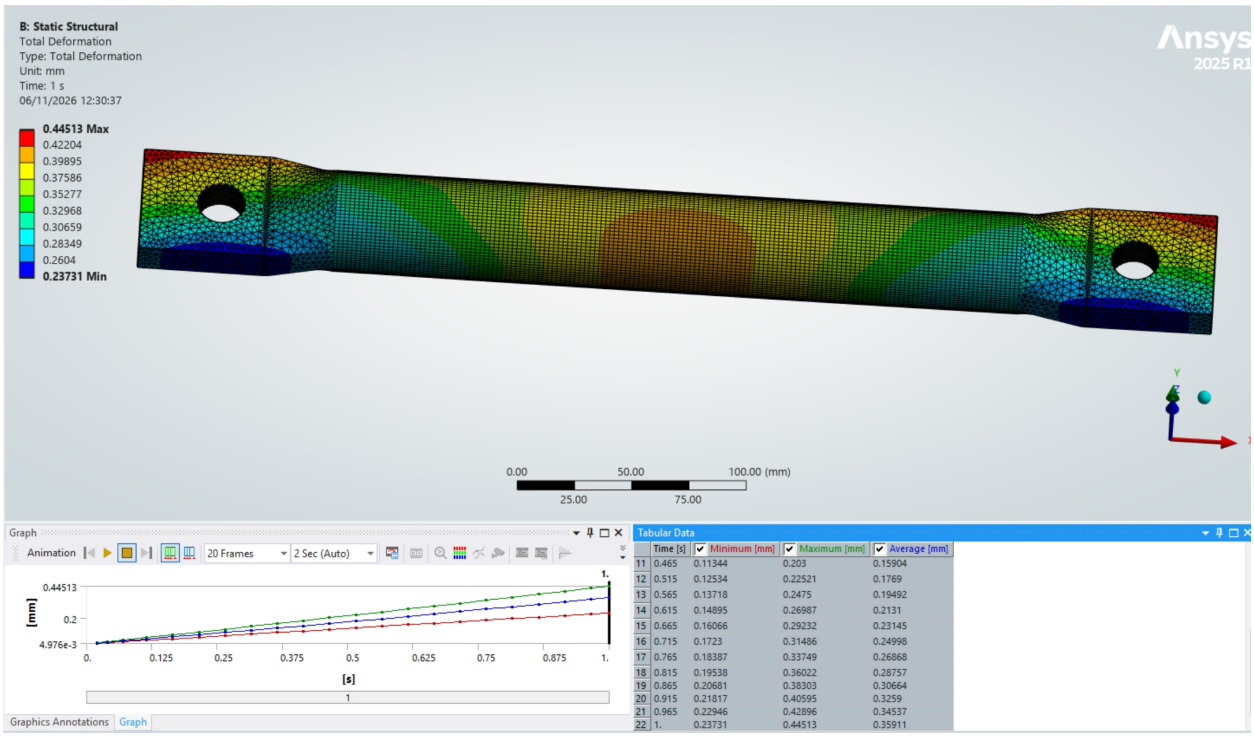
Eccentricity e (mm)	Peak load (kN)	Governing mechanism	Peak / limit load	Lateral deflection at limit load (mm)
0.5	83.2	Johnson load reached before first yield	1.96	0.102
1.0	78.4	First yield	1.85	0.205
2.0	70.2	First yield	1.65	0.410

First yield is defined as the load where secant peak stress reaches the typical yield strength of 276 Mpa. If the strut reaches its column buckling limit beforehand, that critical load governs.

The predicted peak failure load is $70 \text{ to } 83 \text{ kN}$ ($16 \text{ to } 19 \text{ kip}$). As shown in Figure 4, failure is expected to initiate via material yielding on the concave face at mid-length, followed immediately by rapid lateral bowing while all end components remain fully elastic. All proof test loads (42.5, 53.1, and 59.5 kN) sit safely below the lowest predicted first-yield threshold (70.2 kN). At an eccentricity of 1 mm, the first-yield load efficiency is 97 kN/kg.

9. Finite element analysis

The results of the FEA can be viewed in the GitHub folder “ANSYS FEA”.



9.1 Purpose and approach

The hand calculations of Sections 4 to 8 predict how the strut behaves. The finite element analysis (FEA), run in ANSYS Mechanical before any metal is cut, is an independent second calculation with three jobs:

1. Check that stiffness, stress and deflection agree with the hand calculations. Agreement from two different methods is strong evidence that both are right.
2. Find the elastic buckling modes and their loads, to show which way the strut bows and whether local wall buckling is a risk.
3. Reveal local stress concentrations (hole, shoulder, bore end) that a column formula cannot see.

The work follows the structure of NASA-STD-7009, the NASA standard for models and simulations: verification (is the model solved correctly), validation (does it match reality) and uncertainty assessment [28]. The test of Section 11 provides the validation.

Results: Every compared quantity agrees with the hand calculation within 8.5%, and most within 2.5% (Section 9.6). The nonlinear and joint analyses that remain are listed in Section 9.8.

9.2 Model definition

Item	Setting	Reason
Software	ANSYS Mechanical Enterprise 2025 R1	Widely used in aerospace industry
Geometry	The strut of Sections 6 and 7 (56 mm head, 44 x 2.5 mm tube, 15 mm	Eccentricity is applied only in the loading (Section 9.4), so the structure is exactly

Item	Setting	Reason
	tongue, 20 mm pin hole), perfectly symmetric, with no intentional offset or imperfection	the part to be machined
Symmetry	Plane at mid-length (x = 235 mm), where the reactions are read	The geometry, load and C-shaped bow are symmetric about this plane
Material	6061-T6, E = 68,900 MPa, Poisson's ratio 0.33 [8][10]	Same values as the hand calculation
Units	mm, kg, N, s, MPa	Stress comes out directly in MPa
Static solution	Large-deflection static structural	The bowing changes the bending moment (the P-delta effect), as the secant formula does
Element order	Quadratic (midside nodes kept)	A quadratic element represents a linearly varying strain, which is what bending produces
Pins and clevis	Not in this model. The load is applied at the pin line through a remote point to model eccentricity	The column behaviour is isolated here. The pin joint is a separate model (Section 9.8)

9.3 Mesh, the reason for each choice, and convergence

The mesh uses two element types. The tube is meshed with swept hexahedral elements, three through the 2.5 mm wall. The tongue, shoulder and taper are meshed with tetrahedral elements, with finer sizing on the pin-hole surface.

Region	Mesh	Reason
Tube between the tapers (x = 85 to 235 mm)	Swept hexahedral, 3 elements through the wall	The tube is a regular shape that sweeps cleanly along its axis. A swept mesh needs about half the degrees of freedom of a tetrahedral mesh for the same accuracy [30], its elements follow the axis and the wall, and three elements through the wall resolve the bending gradient
Tongue, head, shoulder, taper	Patch-conforming tetrahedral	Flats cut in a round bar, a shoulder, a taper and a hole cannot be swept. A hexahedral mesh is not always possible on complex geometry, and a good higher-order tetrahedral mesh is more accurate than a distorted hexahedral one [29]
Pin hole surface	Face sizing, finer than the surroundings	Stress varies steeply around a hole and the load enters here

Region	Mesh	Reason
Interface at x = 85 mm	Shared nodes between the two bodies	The two bodies form one part, so there is no contact and no artificial stiffness

Mesh convergence. The same case (e = 2 mm, 42.5 kN) was solved on a coarse mesh and on a mesh refined 1.5 times. The acceptance rule is a change of 5% or less.

Output	Coarse mesh	Fine mesh	Change
Maximum total deformation (mm)	0.44513	0.44535	+0.05%
Mid-length stress, concave side (MPa)	159.6	159.9	+0.2%
Mid-length stress, convex side (MPa)	96.2	96.4	+0.2%

The coarse values are equivalent (von Mises) stress and the fine values are normal stress along the axis. At these surface points the stress is uniaxial, so the two are the same quantity. All three outputs are mesh-independent within 0.2%. The peak stress at a corner or contact edge is deliberately not used as an output, because such a stress rises with every refinement (Section 9.6).

9.4 Loads, restraints and how eccentricity is applied

Item	Setting	Reason
Axial load	42.5 kN (the limit load of Section 4)	Same load as the hand calculations
Eccentricity	A remote force located at the pin-hole centre (x = 35 mm) and offset by e in the in-plane direction. Cases e = 0, 0.5, 1.0 and 2.0 mm	A remote force acts like a regular force plus a moment, and its origin can be placed anywhere by entering coordinates [25]
Hinge at the pin	A remote displacement at the same point that restrains lateral motion and twist but leaves rotation about the pin axis free [26]	The pin lets the end rotate in the bending plane
Symmetry plane	Frictionless support on the mid-length face	Enforces zero axial displacement there, which is the symmetry condition

The eccentricity moment at each end is:

$$M_{end} = P\,e$$

This equals 21.3, 42.5 and 85.0 N m for e = 0.5, 1.0 and 2.0 mm at 42.5 kN. It is the end moment that the secant formula of Section 5.3 assumes, so the FEA and the hand model describe the same physical case.

Why the eccentricity is not built into the geometry: a real pin carries only force. The contact pressure on a cylindrical hole is radial, so its resultant passes through the hole centre, and the offset between load line and strut axis is exactly what the remote point represents. Keeping the strut symmetric also means all four cases use the same mesh, so differences between cases come from the load alone.

9.5 Verification of the load path

Check	Expected	FEA	Result
Axial force reaction at the symmetry plane	42.5 kN	42.5 kN	Pass
Moment reaction, e = 2 mm	85.0 N m (P x e)	84,999 N mm = 85.0 N m	Pass
Moment reaction, e = 0, 0.5, 1.0	0, 21.3, 42.5 N m	Equal to P x e in each case	Pass

These are equilibrium checks. They confirm the loads were entered correctly, but a model with the wrong stiffness would pass them too, so the stiffness and stress checks follow in Section 9.6.

9.6 Comparison with the hand calculations

Worked hand calculation (e = 2 mm, 42.5 kN). The tube has A = 325.9 mm2, I = 70,423 mm4, r = 14.70 mm and c = 22 mm. The secant argument is:

$$\varphi = \frac{KL}{2r}\sqrt{\frac{P}{AE}} = \frac{400}{2(14.70)}\sqrt{\frac{42{,}500}{325.9 \times 68{,}900}} = 0.5919, \qquad \sec\varphi = 1.2050$$

The added lateral deflection and the mid-length moment, including the P-delta effect, are:

$$\delta = e(\sec\varphi - 1) = 2(0.2050) = 0.410 \text{ mm},$$

$$M_{mid} = P(e + \delta) = 42{,}500 \times 2.410 = 102.4 \text{ N m}$$

The stresses on the two sides of the mid-length section are:

$$\sigma_{concave} = \frac{P}{A}\left[1 + \frac{e\,c}{r^2}\sec\varphi\right] = 130.4\,[1 + 0.2454] = 162.4 \text{ MPa}$$

$$\sigma_{convex} = 130.4\,[1 - 0.2454] = 98.4 \text{ MPa}$$

The mid-length moment (102.4 N m) is larger than the end moment (85.0 N m) because the bowing adds the lever arm delta. The moment reaction of Section 9.5 is 85.0 N m for any value of delta, because it is measured about the undeformed centroid of the support face, so only the mid-length stress and deflection test the P-delta amplification.

Targets for every case at 42.5 kN (script validation_targets.py, Appendix E):

e (mm)	End moment (N m)	Added deflection at mid-length (mm)	Mid-length moment (N m)	Compressive stress, concave side (MPa)	Compressive stress, convex side (MPa)
0	0	0	0	130.4	130.4
0.5	21.3	0.102	25.6	138.4	122.4
1.0	42.5	0.205	51.2	146.4	114.4
2.0	85.0	0.410	102.4	162.4	98.4

Comparison for the most severe case (e = 2 mm, 42.5 kN). The criteria are decisions: 5% for quantities that do not depend on bending, and 10% for quantities that do, to allow for the hand model's uniform-tube idealisation.

Quantity	Hand calculation	FEA	Difference	Criterion	Result
Mid-length stress, concave side	162.4 MPa	159.9 MPa	-1.6%	10%	Pass
Mid-length stress, convex side	98.4 MPa	96.4 MPa	-2.0%	10%	Pass
Stress on the neutral-axis lines	130.4 MPa (P/A)	about 130 MPa	about 0%	5%	Pass
Added mid-length deflection	0.410 mm	0.400 mm	-2.4%	10%	Pass
Tube shortening over 150 mm	0.284 mm	0.308 mm	+8.5%	10%	Pass (largest difference)

The neutral-axis lines are the lines of the section on which in-plane bending does not act, so the stress there is the plain axial stress P/A. The shortening has the largest difference, and it depends on the line it is measured along: bending adds strain on the concave surface and removes it on the convex one, so the same 150 mm gives about 0.348 mm on the concave surface, 0.209 mm on the convex surface and 0.284 mm on the neutral axis. The FEA value lies inside that range. The like-for-like comparison is the neutral-axis line, which is re-read from the solution before the report is issued.

Where the stress is highest. The equivalent-stress plot shows the highest visible tube stress, 194 MPa, at the junction of the taper and the tube, against 160 MPa at mid-length. The junction is where the thick solid section meets the 2.5 mm wall, so a stress rise there is expected. At 1.25 times the limit load, and with e = 2 mm, this value scales to about 243 MPa, close to the 240 MPa design yield strength. The junction is therefore the likely place of first local yield, and it is where the shoulder strain gauges of Section 11.4 sit.

9.7 Linear buckling results

The eigenvalue buckling analysis uses the e = 2 mm static solution as pre-stress. The load multiplier is the factor by which the pre-stress state, force and moment together, must grow to cause buckling [27]. The table gives the modes that were examined on the coarse mesh. The equivalent axial stress is the multiplier times P/A = 130.4 MPa.

Mode	Load multiplier	Critical load (kN)	Shape	Comparison with hand calculation
1	5.949	252.8	Bow of the whole strut in the plane where the end can rotate	Euler load for pinned ends is 299.3 kN (multiplier 7.04). FEA is 15.5% lower, an effective length factor of

Mode	Load multiplier	Critical load (kN)	Shape	Comparison with hand calculation
				1.09
2	11.92	506.5	Bow of the whole strut in the plane the forks restrain	2.0 x mode 1. Effective length factor 0.77, between pinned (1.0) and fixed-pinned (0.7) [11]
3	24.46	1,040	Three-lobed ripples in the tube wall	Equivalent stress 3,190 MPa, 0.64 x the classical 5,022 MPa [16][24]
5	27.87	1,184	Four-lobed ripples in the tube wall	3,634 MPa, 0.72 x classical
7	28.69	1,219	Local buckling concentrated near both tapers	3,741 MPa, 0.75 x classical
10	29.79	1,266	Ripples along the whole tube	3,884 MPa, 0.77 x classical

Result Interpretation

1. **Elastic buckling is not what limits the strut.** The first mode needs 253 kN, three times the predicted peak load of 83 kN. The strut yields first, which is the stocky-column behaviour of the Johnson regime (Section 5.2). A nonlinear analysis with yielding (Section 9.8) predicts the peak load.
2. **The forks act as partial restraints.** The second whole-strut mode, in the plane the forks hold, needs twice the load of the first. The bow in the free plane governs, so an eccentricity in the pin plane (Section 9.4) is the critical case.
3. **Local wall buckling is not a risk.** The lowest local multiplier is 24.5, an axial stress of 3,190 MPa, 13 times the design yield strength. The classical formula gives a higher value, as expected, because it describes an ideal cylinder of infinite length.
4. **Mode 1 is 15.5% below the Euler load.** Two effects contribute. The pre-stress contains the bending of the e = 2 mm case, and the multiplier scales force and moment together. The pinned end is also not an ideal hinge, because it is held at a deformable remote point. Running the same analysis with e = 0 separates the two effects. It should return a multiplier close to 7.0 (Section 9.8).
5. **The local-buckling multiplier is a useful cross-check of the load.** A multiplier near 30 corresponds to an axial stress near 3,900 MPa only if the pre-stress is close to 130 MPa, the hand-calculated P/A. This supports the loading in the buckling solution.

9.8 Findings and credibility

6. The hand model is verified: stress, deflection and stiffness agree within 2.4%, and the shortening within 8.5%, for the worst eccentricity.

7. The result does not depend on the mesh: 0.05% to 0.2% between two meshes.
8. Elastic buckling needs three times the predicted peak load, so yielding governs, as the Johnson calculation of Section 5.2 assumed.
9. Local wall buckling is more than an order of magnitude above yield.
10. The forks restrain the out-of-plane bow, so the in-plane eccentricity is the critical case.

Credibility based on NASA-STD-7009 standards [28].

Aspect	Evidence in this project	Status
Verification of the solution	Equilibrium checks; agreement with hand calculations within 8.5%; mesh comparison within 0.2%; singularity identified and not reported	Complete for the static and buckling cases
Validation	Test of Section 11 compared with the hand calculation and the FEA	After the test
Input pedigree	Material from mill certificate and coupon tests; dimensions measured	After machining
Uncertainty	Sensitivity to eccentricity (four cases), friction, wall thickness and strength	Eccentricity done; the others are runs 5 and 6
Limitations	Pin joint modelled separately from the column; tension-lug coefficients are placeholders	Stated in Section 15

10. Fabrication plan

10.1 Material and stock

11. Material: 6061-T6 round bar, 56 mm diameter, enough for two struts (one test article and one spare), each 470 mm long plus cutting allowance.
12. Request the mill certificate with the guaranteed minimum yield and ultimate strength, and compare with the design minimums of 240 and 290 MPa [10]. Cut a tensile coupon from the same bar for each strut (Section 11.4).
13. Ask the supplier whether a stress-relieved temper is available, because boring a 39 mm hole releases stress in the bar and can bend it. The supplier's answer decides the sequence in Section 10.2.

10.2 Process route

1. Saw the bar 10 mm over length.
2. Rough-turn the outside diameter, leaving 0.5 mm on the diameter.

3. Bore from both ends toward the middle, 150 mm deep each, to a 38.5 mm diameter. Boring from both ends avoids a single 300 mm deep bore (a length-to-diameter ratio of 7.7). The two bores meet at mid-length.
4. Finish-bore to 39.0 mm and finish-turn the outside to 44.0 mm over the tube length, the two 30 mm tapers and the 56 mm heads, with the part held between centres so that bore and outside are concentric within 0.1 mm.
5. Mill the two flats at each end to the 15.0 mm tongue thickness, with the R5 shoulder fillet (Section 10.5).
6. Drill and ream the two pin holes to 20 mm H7, in one setup, with the hole centres 400.00 mm apart.
7. Break all edges 0.5 mm and chamfer the hole edges 0.5 mm.
8. Mark each strut, inspect it (Section 10.3), and keep the inspection record.

10.3 Critical features and inspection plan

Feature	Why it matters	Method	Tolerance
Wall thickness	Sets the strut capacity: the Johnson ceiling changes by about 3 kN for each 0.1 mm of wall	Ultrasonic thickness gauge at 8 points around the circumference at 3 stations, plus a bore gauge at both ends	2.5 mm plus or minus 0.1 mm
Bore and outside concentricity	A thin wall on one side raises the eccentricity of the load path	Dial indicator with the part on V-blocks, or CMM	0.1 mm
Pin-hole centre distance	Sets the column length L	CMM, or a calibrated pin gauge and micrometre	400.00 mm plus or minus 0.02 mm
Pin-hole size	Controls the pin clearance	Go and no-go plug gauges	20 H7 (20.000 to 20.021 mm)
Hole offset from the strut axis	Directly adds to the eccentricity	CMM	plus or minus 0.05 mm
Straightness	An initial bow behaves like an eccentricity	Dial indicator on V-blocks while rotating	0.1 mm over the tube length (decision)
Tongue thickness	Sets the net section and the pin bending arm	Micrometre at four points	15.0 mm plus or minus 0.1 mm

10.4 Pins, fit and pin retention

What the fit codes mean. The ISO 286 system describes a size limit with a letter and a number. A capital letter is a hole and a lowercase letter is a shaft. The letter gives the position of the tolerance zone relative to the nominal size: H starts exactly at the nominal size and goes upward, and g sits slightly below it. The number is the tolerance grade, which sets the width of the zone: for sizes of 18 to 30 mm, grade 7 is 21 micrometres wide and grade 6 is 13 micrometres wide. The hole 20 H7 therefore measures 20.000 to 20.021 mm and the pin 20 g6 measures 19.980 to 19.993 mm, so the clearance is 7 to 41 micrometres [19][20]. This is a sliding fit: the pin locates the strut accurately and still turns freely.

Pin specification. 20 g6 x 48 mm, ground, 4130 steel, specified by minimum properties and not only by name: tensile strength of at least 862 MPa and yield strength of at least 710 MPa, the values used in the capacity checks [14]. Ends chamfered 1 mm x 45 degrees to let the circlips slide on. The fork plates use the same steel and strength minimums. The pin surface is lubricated (Section 11.3).

Circlip groove design procedure. A circlip (retaining ring) holds the pin in place. It does not carry the test load.

1. **Choose the standard.** External retaining rings to DIN 471, a common metric standard sold by general suppliers [31][32].
2. **Read the ring and groove sizes for a 20 mm shaft.** Ring thickness 1.20 mm, groove diameter 19.0 mm, groove width 1.30 mm [31]. The supplier's groove width tolerance is +0.14 mm and 0, and the groove bottom radius must not exceed 0.127 mm [32].
3. **Set the groove diameter tolerance.** 19.0 mm, 0 to minus 0.1 mm (my decision, inside the ring supplier's tolerance band). A groove that is too deep weakens the pin, and one that is too shallow lets the ring spin out.
4. **Find the stack that sets the groove positions.** Fork 10 + gap 0.5 + tongue 15 + gap 0.5 + fork 10 = 36 mm between the outer fork faces. The pin is 48 mm long, so 6 mm projects beyond each fork face.
5. **Place the groove.** The ring sits against the groove's inner wall. Put the inner wall 5.8 mm from the pin end, so the ring is 0.2 mm from the fork face (a small axial float so that the ring does not press on the fork). The groove is 1.30 mm wide, so its outer wall is 4.5 mm from the end. Check that this edge distance exceeds the minimum edge distance in the DIN 471 table before ordering.
6. **Check the space for the ring.** The ring needs a free outside diameter of 28.4 mm [31]. The fork plates are flat at the pin, so the ring has room.
7. **Write the groove callout on the pin drawing.** 19.0 mm diameter, 0 to minus 0.1; width 1.30 mm, plus 0.14 and 0; groove outer wall 4.5 mm from each end; bottom radius 0.12 mm maximum; surface finish Ra 1.6 micrometres (a smooth finish lowers the stress at the groove).
8. **Install.** Use circlip pliers. Check that each ring is fully seated and rotates freely.

Why the retention is not a structural concern. The pin carries no axial load in the test. The rings only prevent the pin from walking out under vibration or handling.

10.5 Specimen drawing details

The table of tolerances is in Section 11.5. Two details complete the CAD model:

- **Shoulder fillet R5.** A fillet where each flat meets the shoulder face removes a sharp internal corner. A larger radius would reach the fork plate nose.
- **Fork plates end at x = 48 mm.** The fork plates stop 2 mm before the shoulder fillet begins, because the 56 mm head fills the space beyond. The ligament beyond the hole is then only 3 mm. This is acceptable because in compression the pin loads the base side of the fork holes. The clevis cannot be used for a tension test.

10.6 Manufacturing risks

Risk	Effect	Mitigation
The two bores do not meet or differ in diameter	Wall thickness steps at mid-length	Bore from both ends with the part held in one setup per end, measure wall thickness at mid-length, and reject a step above 0.1 mm
The wall is off-centre	Adds eccentricity	Concentricity check (Section 10.3)
The bar bends when the stress is released	Initial bow	Rough machine, rest, then finish; straightness check
Hole position error	Adds eccentricity	Ream both holes in one setup and measure on a CMM
Tongue flats not parallel to the pin axis	Uneven bearing on the pin	Mill both flats in one setup and check parallelism

11. Test plan

11.1 Facility and machine capacity

The planning load is 1.5 times the predicted peak: 1.5 x 83.2 = 125 kN (28 kip, using 1 kip = 4.448 kN). The USC Structures and Materials Research Laboratory [15] lists universal testing machines from 1 to 1,000 kip and servo-hydraulic actuators from 11 to 340 kip, with instrumentation for load, displacement and strain. The plan assumes a machine rated at least 50 kip (222 kN), so the expected load sits well inside the load-cell range. Before machining, confirm with the laboratory:

- the machine and load cell, and the load-cell accuracy between 20 and 120 kN;
- the vertical space: a 470 mm strut plus two clevises plus the load train needs about 650 mm or more;
- the grip or adapter interface, which defines the spigot of the clevis (the CAD spigot is a placeholder);
- the number of data-acquisition channels (at least 10: four mid-length gauges, four shoulder gauges, the lateral sensor and the extensometer), and whether strain-gauge amplifiers are available;
- access rules and the training that a student needs.

11.2 Fixture

- Two steel fork plates per end, 10 mm thick, with 20 H7 holes and a gap of at most 0.5 mm per side. Material and strength minimums as for the pins (Section 10.4).
- Pin centres define the 400 mm length, and the load passes through the pins, not through flat platens, so the pinned end condition of the model is preserved.
- **Fork check at the planning load.** Each plate carries half the load, 62.5 kN. Its bearing capacity from the Air Force manual forms with the 4130 strengths is min(1.33 x 861.8, 1.304 x 1.33 x 710.2) x 20 x 10 = 229 kN, a margin of +2.67. The net-section capacity through the hole is 0.74 x 861.8 x (56 - 20) x 10 = 230 kN, also +2.67 [14].
- Fixture axial stiffness of at least 620 kN/mm, ten times the predicted strut stiffness of 61.9 kN/mm, so that the fixture does not change the end condition.
- The pins are designed to stay elastic to 91.5 kN and have an elastic limit of 121 kN. The test is stopped and the pins are inspected if the load exceeds 100 kN without a peak.
- A guard surrounds the specimen, because column failure releases stored energy suddenly.

11.3 Load eccentricity and pin friction

Two effects shift the line of action of the load relative to the strut axis, and the FEA treats both as the eccentricity e.

Alignment. A tolerance-stack estimate gives hole-position error 0.05 mm, fork and fixture misalignment about 0.2 mm and pin clearance at most 0.02 mm (half of 41 micrometres). The worst-case sum is 0.27 mm and the root-sum-square is 0.21 mm, so the analysis value of 1.0 mm is conservative by a factor of four to five. The tested range of 0.5 to 2.0 mm brackets it.

Pin friction. A pin that turns in its hole under load P does not pass the load through its centre. The resultant is tangent to a friction circle of radius r sin(phi), where tan(phi) equals the friction coefficient, which is close to mu times r [34]. For the 20 mm pin (r = 10 mm):

Friction coefficient	Friction-circle radius (mm)	Friction moment at 42.5 kN (N m)	Typical condition
0.1	1.0	42	Well lubricated
0.2	2.0	83	Greased metal pair (0.16 to 0.19 for steel, brass and copper on steel) [33]
0.47	4.3	181	Dry aluminium on mild steel, kinetic (static is 0.61) [33]

The friction moment has the same size as the eccentricity moment (21 to 85 N m for e = 0.5 to 2 mm). Friction opposes rotation, so it acts as a partial end restraint that stiffens the strut and delays bowing, and it hides the true eccentricity. **Decision:** grease the pins with a light grease, record the lubricant and the quantity, and use a friction coefficient of 0.2 as the base case in the joint model, with 0.1 and 0.5 as bounds.

Measuring e. For a pair of gauges on opposite sides of the tube with strains e1 and e2:

$$\varepsilon_{bend} = \frac{\varepsilon_1 - \varepsilon_2}{2}, \qquad M = \frac{E\,I\,\varepsilon_{bend}}{c}, \qquad e = \frac{M}{P\,\sec\varphi}$$

with E = 68,900 MPa, I = 70,423 mm4 and c = 22 mm. The moment at mid-length is P(e + delta) = P e sec(phi), so dividing by P sec(phi) gives e. Read the strains at a low load (about 10 kN), where the secant term is close to 1.04. A second estimate comes from fitting the secant curve to the measured load-deflection data.

11.4 Pre-test measurement and instrumentation

Before the test, measure and record:

- the tube outside diameter and wall thickness at both ends and at mid-length (Section 10.3);
- the straightness, the hole positions and the hole diameters;
- the yield and ultimate strength of the tensile coupons cut from the same bar, so that the typical strengths used in the predictions are replaced by measured ones.

Instrumentation:

- four strain gauges at mid-length, 90 degrees apart around the tube, wired so that opposite pairs give the bending directly (the axial strain cancels) and the sum gives the axial strain;
- two gauges at each taper-to-tube shoulder, to watch the transition;
- a lateral displacement sensor at mid-length, in the pin plane;
- an axial extensometer between the pin centres, which keeps machine compliance out of the stiffness measurement;
- the machine load cell and video of the specimen;
- a data rate of at least 10 samples per second (decision).

11.5 Specimen tolerances

These tolerances are for a first machined part and are confirmed with the machine shop.

Feature	Nominal	Tolerance or note
Material	6061-T6 bar, 56 mm	Mill certificate required
Overall length	470 mm	plus or minus 0.2 mm
Pin-hole centre distance	400.00 mm	plus or minus 0.02 mm
Pin holes	20 mm	H7 (20.000 to 20.021 mm), Ra 1.6 micrometres, perpendicular to the tongue faces within 0.02 mm
Hole centre offset from strut axis	0	plus or minus 0.05 mm
Tongue thickness (milled flats)	15.0 mm	plus or minus 0.1 mm

Feature	Nominal	Tolerance or note
Tongue zone length	55 mm	plus or minus 0.2 mm
Edge distance e_h	35 mm	plus or minus 0.1 mm
Shoulder fillet	R5	smooth, no tool marks transverse to the axis
Solid taper	30 mm long, 56 mm to 44 mm	smooth
Tube outside diameter	44.0 mm	plus or minus 0.1 mm
Tube bore	39.0 mm, 300 mm long	bored from both ends, meeting at mid-length; wall 2.5 mm plus or minus 0.1 mm
Concentricity of bore and outside diameter		0.1 mm
Straightness over the tube length		0.1 mm
Edges	break 0.5 mm	removes edge starters

11.6 Test procedure

- Confirm the inspection record and the coupon results. Enter the measured wall, hole positions and yield strength into the hand calculation and the FEA. These become the post-test predictions (Section 12.1). The pre-test predictions of Section 8 stay unchanged.
- Grease the pins (Section 11.3), fit the circlips and assemble the specimen in the two clevises.
- Connect the gauges and record the zero-load strains. They show the assembly misalignment.
- Mount the assembly in the machine and align the spigots with a dial indicator. Apply a 2 kN seating load and compute e from the gauges. If e exceeds 2 mm, unload, correct the shimming of the fork gap and repeat.
- **Stage 1.** In displacement control, at 0.5 mm/min (decision, slow enough to be quasi-static), load to 42.5 kN, hold 30 s, unload to 2 kN, and record any residual strain or lateral offset.
- **Stage 2.** Load to 53.1 kN (1.25 x limit), hold 30 s, unload and check for permanent set. This verifies requirement R1.
- **Stage 3.** Load to 59.5 kN (1.4 x limit), hold 60 s, unload. This verifies requirement R2. All three stages are below the lowest predicted first yield of 70.2 kN, so they should not damage the part.
- **Stage 4.** Continue at the same rate until the load peaks and then falls by 20%, or until the lateral deflection reaches 10 mm (decision), and then stop.
- Unload, photograph the specimen, and measure the residual bow.
- Inspect the pins (straightness and diameter), the tongue holes (ovalisation, with the go and no-go gauges) and the fork plates. Section the tube at the yield location to measure the wall.

11.7 Data processing

- Convert the load cell and gauge channels to axial load, axial strain, bending strain in each plane and mid-length lateral deflection.
- Axial stiffness: the slope of load against extensometer shortening in the elastic range. The bar-model prediction is 61.9 kN/mm. The measured value will be lower because the pin, hole and fork compliance are not in the bar model. Report both.
- Eccentricity: from the low-load gauge readings and from a secant fit to the load-deflection curve (Section 11.3).
- First yield: the load at which the concave-side strain departs from the elastic line, or at which the strain equals the yield strain of the coupon.
- Peak load, load after the peak, and the position of the bow.

11.8 Verification matrix

ID	Requirement	Verification	Predicted margin (hand)	FEA check
R1	Yield at 1.25 x limit	Analysis; test shows no permanent set after unloading from 53.1 kN	+0.30	Mid-length stress within 2% of hand. Highest tube stress, 194 MPa at the taper junction (e = 2 mm, limit load), is tracked by the shoulder gauges
R2	Ultimate at 1.4 x limit	Analysis; 60 s hold at 59.5 kN	+0.40	Same
R3	Buckling margin of at least 10%	Analysis; eigenvalue FEA	+0.12	Mode 1 at 252.8 kN, 3.0 x the predicted peak and 3.9 x the 65.5 kN required
R4	Wall D/t at most 20, t at least 2.0 mm, local buckling stress at least 3 x yield	Analysis; FEA; measured wall	D/t = 17.6; local stress 20.9 x yield	Lowest local mode at 3,190 MPa equivalent, 13 x design yield
R5	Ends at least 1.25 x strut maximum; pins elastic to 1.10 x	Analysis; joint model; post-test inspection	+0.05 (tongue net section, the tightest); +0.32 (pin elastic)	Joint model planned (Section 9.8)
R6	Axial stiffness measured	Extensometer between pin centres	Predicted 61.9 kN/mm (bar model)	Tube shortening within 8.5% of hand
R7	Peak load within 15%	Correlation table	To be	Nonlinear run planned

ID	Requirement	Verification	Predicted margin (hand)	FEA check
	of the prediction made with measured inputs	(Section 12.1)	determined by the test	(Section 9.8)

12. Correlation and redesign

12.1 Correlation method

Three predictions are compared with the measurement, which separates input uncertainty from model error:

- **P1, before the test:** hand calculation with typical strength and e = 1 mm (Section 8): first yield 78.4 kN, ceiling 83.2 kN. These numbers are fixed and are not changed after the test.
- **P2, after measurement:** the same hand calculation with the measured wall, coupon strength and eccentricity.
- **P3, the FEA:** the nonlinear run of Section 9.8 with the same measured inputs.

For each row of the table the percent error is (measured - predicted) / predicted.

Quantity	P1 hand (typical)	P2 hand (measured inputs)	P3 FEA	Measured	Error
Axial stiffness (kN/mm)	61.9	to be completed	to be completed	to be completed	
Eccentricity e (mm)	1.0 assumed	from gauges	cases 0, 0.5, 1.0, 2.0 run	from gauges	
Added lateral deflection at 42.5 kN, e = 2 mm (mm)	0.410	to be completed	0.400	to be completed	
Mid-length stress at 42.5 kN, concave side, e = 2 mm (MPa)	162.4	to be completed	159.9	from gauges	
First yield (kN)	78.4 (e = 1 mm)	to be completed	to be completed (nonlinear run)	to be completed	
Peak load (kN)	70.2 to 83.2	to be completed	to be completed (nonlinear run)	to be completed	

Quantity	P1 hand (typical)	P2 hand (measured inputs)	P3 FEA	Measured	Error
Failure mode and location	Yield at the concave face, then bow				match or not

The criterion R7 is that the peak load agrees within 15%. Two outcomes are equally informative: agreement shows that the models are good enough to size a flight strut without testing every iteration, and a disagreement with a quantified cause shows which input or assumption to fix.

12.2 Diagnosis of the observed failure

Observation	Most likely cause	Design response
Sudden bow near 83 kN with little prior yielding	Stability governs	Increase the radius of gyration per unit mass or reduce the effective length
Peak between 70 and 83 kN with yielding on the concave face	Eccentricity-driven yield, as predicted	Reduce the eccentricity with tighter tolerances, lubrication control or spherical end bearings
Peak below 70 kN	Eccentricity above 2 mm, strength below typical, or a thin wall	Check the measured e, coupon strength and wall, and correct the model inputs
Peak above 83 kN	Pin friction restraining the end rotation, or strength above typical	Compare the measured e with the friction estimate of Section 11.3; repeat with different lubrication
Fracture or ovalisation of a tongue below 105 kN	Tension-lug coefficients K and Kn were optimistic	Increase the tongue thickness or width; refine K and Kn from the manual's figures
Permanent set in a pin	Pin strength lower than assumed, or bending arm larger than modelled	Use a larger or stronger pin; check the arm
Wall dimple near mid-length	Local buckling worse than the FEA predicts	Increase wall thickness; check the measured wall
Crack at the taper shoulder	Stress concentration at the shoulder	Increase the shoulder radius

12.3 Redesign levers and success metric

The first-yield load per unit mass at e = 1 mm is 78.4 kN / 0.805 kg = 97 kN/kg. The end regions are 67% of the part mass, and the shank mass changes little with eccentricity up to 2 mm (Section 6.2), so the main mass reductions lie in the end regions: a shorter tongue zone and taper, pocketed tongue faces, and a thinner tongue with a bushing. A second lever is a spherical end bearing, which removes the friction and alignment effects of Section 11.3 and moves the first-yield load toward the Johnson ceiling. The success criterion for the next revision is an increase of at least 30% in peak load per unit mass with every requirement of Section 3.2 still met. One change is made at a time, so that each gain can be attributed.

13. Project plan, risks and open items

13.1 Plan

The durations are estimates and depend on laboratory access and machine-shop lead time.

Step	Activity	Output	Status
1	Static FEA for e = 0, 0.5, 1.0, 2.0 mm; mesh comparison; eigenvalue buckling	Sections 9.3 to 9.7	Done
2	Bore-bottom radius study; nonlinear run to peak load; joint model with contact; buckling with e = 0	Section 9.8 items 1 to 5	Weeks 1 and 2
3	Confirm test machine, grips, channels and access with the laboratory; final drawings including the bore-bottom radius	Fixture interface fixed; drawing set	Week 2
4	Order bar stock, pins, circlips and strain gauges	Purchase orders	Week 3
5	Machine two struts and the fixture	Parts	Weeks 4 to 6
6	Inspection and coupon tests	Inspection record, measured strength	Week 7
7	Gauge installation; update the hand calculation and FEA with measured inputs; rehearsal with the fixture and a dummy	Post-measurement predictions	Week 8
8	Test	Data	Week 9
9	Correlation, redesign recommendation, final report	Section 12.1 table complete	Week 10

13.2 Risks

Risk	Effect	Mitigation
Local stress at the taper	First yield below the	Corner-radius study; shoulder strain gauges;

Risk	Effect	Mitigation
junction or the bore bottom starts yielding early	predicted range	hand and FEA first-yield definitions compared with the strain data
Nonlinear FEA disagrees with the hand prediction by more than 10%	Peak-load prediction uncertain	Review inputs (boundary stiffness, material curve); report both predictions; the test decides
Machining misses the wall tolerance	Strut capacity differs from the prediction by about 3 kN per 0.1 mm	Inspect, use the measured wall in the post-test prediction, machine two struts
Eccentricity is larger than 2 mm	Peak load below the predicted range	Alignment check at the 2 kN seating load; correct before loading
Pins yield before the strut	Test corrupted	Stop at 100 kN; spare pins
Laboratory access or machine unavailable	Schedule slip	Confirm early; book the machine in week 2
Gauge or sensor failure	Loss of the eccentricity measurement	Spare gauges; redundant extensometer and lateral sensor
Bar stock lead time	Schedule slip	Order in week 3, with the mill certificate requested

13.3 Open items

- Choose the bore-bottom corner radius from the axisymmetric study (Section 9.8) and add it to the drawing table of Section 11.5.
- Re-read the tube shortening on the neutral-axis line, and complete the buckling runs with e = 0 and on the fine mesh.
- Run the nonlinear peak-load analysis and the joint model (Section 9.8).
- Replace the tension-lug placeholders (net-tension coefficient 0.74 and bearing coefficient 1.33) with the values read from the Air Force manual's figures [14], and the 4130 steel strengths of the manual's example with the certificate values of the pins and forks.
- Confirm the circlip edge distance against the DIN 471 table (Section 10.4), and confirm the machine, grip interface, data channels and vertical space with the laboratory.
- Decide whether to test a tension case. The clevis here is for compression only.

14. Extensions

- **Combined axial and lateral load.** Gimbaling adds a lateral load of about T sin(delta) at the engine, and engine inertia adds more. A combined-load strut and a beam-column analysis are the natural next step.

- **Fatigue and reuse.** Reusable vehicles see many load cycles. The MatWeb extract [9] lists a 6061-T6 fatigue strength of 96.5 MPa at 500 million fully reversed cycles, a starting point for a life estimate with scatter.
- **Low temperature.** MatMatch [18] lists 6061-T6 properties at -196 C, which are higher in strength than at room temperature. A cold test needs a chamber that covers the temperature.
- **Dynamic stiffness.** A tap test of the free-free strut and of the loaded strut gives the first bending frequencies, which compare with the modal analysis (Section 9.8). NASA SP-8055 [17] describes why structural frequencies must stay clear of propulsion-system frequencies, and an early thrust-mount paper [21] discusses the stiffness requirements of thrust mounts.
- **Probabilistic study.** A Monte Carlo analysis that randomises eccentricity, strength, wall and friction gives a failure-load distribution to compare with the measurement.
- **Full thrust structure.** A four-strut assembly with a thrust ring, including the case of one strut failing, tests load redistribution.
- **Tension case.** The lug equations are formulated for tension (Section 5.6; lug failure modes are described in [22]). A tension test would calibrate the K and Kn coefficients directly.

15. Conclusions

- The load on a strut is derived from engine thrust, strut count and strut angle: 224.8 kN at full scale and 42.5 kN on the 0.43-scale test article. The ultimate load is 59.5 kN.
- A two-variable sweep gives a tube of 44.0 mm outside diameter and 2.5 mm wall, 36% lighter than the lightest solid round bar meeting the same criteria. Buckling governs for eccentricities up to 2 mm.
- With the end fittings the part has a mass of 0.805 kg, of which the end regions are 67%. The tongues, holes and pins are sized so that the column fails first. The predicted peak load is 70 to 83 kN, with yielding on the concave face at mid-length followed by lateral bowing.
- The finite element model reproduces the loading exactly (reaction force 42.5 kN, moment reaction 85.0 N m at e = 2 mm). For that worst case it agrees with the hand calculation within 2.4% on mid-length stress and deflection and within 8.5% on tube shortening, and its results change by 0.2% or less between two meshes.
- The first elastic buckling load is 253 kN, three times the predicted peak, so yielding governs. The mode in the plane the forks restrain needs 507 kN, and local wall buckling starts at 24.5 times the limit load, 13 times the design yield strength.
- Two local stress features were found: 194 MPa at the taper-to-tube junction, and a mesh-dependent peak at the bore bottom that is a singularity and not a result. A corner-radius study resolves the second. Pin friction acts like an eccentricity of mu times the pin radius, 1 to 4 mm, so the test controls and records the pin lubrication.
- The fabrication plan, inspection plan, test procedure, correlation table and diagnosis table are defined before machining, so the comparison of hand calculation, FEA and measurement answers the research question directly.

Limitations. The load case is axial thrust only. The strut count, angle and scale are representative values, not vehicle data. The lug coefficients K and Kn are conservative placeholders to be replaced from the manual's figures, and the pin strengths are the manual's example values. The tension lug equations are applied to a compression test as a conservative bound. The FEA models the column and the pin joint

separately, and the nonlinear peak-load, joint and bore-bottom analyses of Section 9.8 are still to be completed. No test results exist yet, so the correlation table of Section 12.1 is completed after the test.

16. Attachments and reproduction

All numbers in this report are produced by the five scripts in Appendices A to E. Run them in this order in one directory: model.py (imported by the others), sweep_tube.py (sizing tables), design_and_prediction.py (design, margins, mass, stiffness and predictions; writes design.json), figures.py (reads design.json and writes the four figures) and validation_targets.py (hand-calculation targets of Section 9 and the comparison with the eigenvalue results).

File	Content
model.py	Material data, load definition, tube section, column, secant, shell-buckling, lug and pin functions
sweep_tube.py	Two-variable sizing sweep and sensitivity tables
design_and_prediction.py	Selected shank, end-fitting search, margins, mass, stiffness and test predictions
figures.py	Figures 1 to 4
validation_targets.py	Hand-calculation targets for the FEA, buckling comparison and modal targets
fig1_free_body_diagrams.png	Load path, eccentric strut and pin-joint free-body diagrams
fig2_geometry.png	Strut geometry
fig3_sizing_map.png	Feasible region and active criteria
fig4_prediction.png	Predicted load against lateral offset

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Appendix A. model.py

```
"""
Common model: 6061-T6 tubular strut with integral clevis-tongue ends, loaded in axial
compression through pins.
Units: N, mm, MPa, kg.  Imported by the sweep, end-fitting and prediction scripts.
"""
import numpy as np

# ---------- material 6061-T6 ----------
E, G, rho = 68.9e3, 26.0e3, 2.70e-6           # MPa, MPa, kg/mm^3
Fty_d, Ftu_d = 240.0, 290.0                   # design (minimum) yield / ultimate,
MPa
Fty_t, Ftu_t = 276.0, 310.0                   # typical values (used for test
prediction only)
```

```
Fsu, Fbru, Fbry = 207.0, 607.0, 386.0          # shear; bearing ultimate / yield (e/D
= 2.0)
# ---------- pin steel (4130 values from Air Force Stress Manual worked example)
----------
ksi = 6.89476
Ftu_P, Fty_P, Fsu_P = 125*ksi, 103*ksi, 82*ksi
kbP = 1.56                                      # plastic bending coefficient,
ductile pin
# ---------- factors and criteria ----------
FS_y, FS_u, MS_buck_min = 1.25, 1.40, 0.10
K = 1.0                                         # pinned-pinned effective length
factor
L = 400.0                                       # pin-centre to pin-centre length, mm
e_ecc = 1.0                                     # load eccentricity used in analysis,
mm
# ---------- load definition ----------
T_full, D_eng = 845e3, 920.0                    # Merlin 1D sea-level thrust (N) and
diameter (mm)
N_str, alpha = 4, np.radians(20.0)
lam = L/D_eng
P_full = T_full/(N_str*np.cos(alpha))
P_L = lam**2*P_full                             # limit load on the test strut
P_y, P_u = FS_y*P_L, FS_u*P_L

# ---------- round tube section ----------
def tube(D, t):
    d = D - 2*t
    A = np.pi/4*(D**2 - d**2); I = np.pi/64*(D**4 - d**4); r = np.sqrt(I/A)
    return A, I, r, D/2

# ---------- column models ----------
def Cc(Fty): return np.pi*np.sqrt(2*E/Fty)
def P_crit(A, r, Fty=Fty_d):
    S = K*L/r
    sig = np.where(S >= Cc(Fty), np.pi**2*E/S**2, Fty - Fty**2/(4*np.pi**2*E)*S**2)
    return sig*A, S
def phi(P, A, r): return K*L/(2*r)*np.sqrt(P/(A*E))
def sigma_secant(P, A, r, c, e=e_ecc):
    f = phi(P, A, r)
    return np.where(f < np.pi/2, P/A*(1 + e*c/r**2/np.cos(np.minimum(f, 1.55))),
np.inf)
def delta_secant(P, A, r, e=e_ecc):
    f = phi(P, A, r)
    return np.where(f < np.pi/2, e*(1/np.cos(np.minimum(f, 1.55)) - 1), np.inf)
def sigma_local(D, t):                          # classical axial shell buckling
stress, 0.605 E t / R (R = mean radius)
    return 0.605*E*t/((D - t)/2)
def first_yield(A, r, c, e, Fty=Fty_t):
    Pc, _ = P_crit(A, r, Fty); Ps = np.linspace(1, 0.9999*Pc, 60000)
    sg = sigma_secant(Ps, A, r, c, e)
    return Ps[np.argmax(sg >= Fty)] if np.any(sg >= Fty) else Ps[-1]     # no yield
before the Johnson load -> ceiling

# ---------- lug and pin (Air Force Stress Manual, double shear, axial) ----------
def pin_capacity(Dp, t_fork, t_tongue, g):
```

```
    arm = t_fork/2 + t_tongue/4 + g                          # eq. 9-13
    return dict(arm=arm,
                ult_bend = 0.1963*kbP*Dp**3*Ftu_P/arm,       # eq. 9-15
                elastic  = 0.1963*Dp**3*Fty_P/arm,
                shear    = 1.571*Dp**2*Fsu_P)                # eq. 9-12
def tongue_capacity(Dp, e, w, t, Kn=0.74, K_b=1.33, Ftu=Ftu_d, Fty=Fty_d):
    Fb = min(K_b*Ftu, 1.304*K_b*Fty, Fbru, 1.304*Fbry) if e/Dp >= 1.5 else np.nan
    return dict(bearing=Fb*Dp*t, net=Kn*Ftu*(w - Dp)*t)
```

Appendix B. sweep_tube.py

```
"""Two-variable sizing sweep (outside diameter D, wall t) for the round tubular strut
shank."""
import numpy as np, json
from model import *

Ds = np.arange(28.0, 60.01, 0.5); ts = np.arange(1.5, 12.01, 0.25)
Dg, tg = np.meshgrid(Ds, ts, indexing="ij")

def evaluate(D, t, e=e_ecc, D_over_t_max=25.0, t_min=2.0):
    A, I, r, c = tube(D, t)
    u_y = sigma_secant(P_y, A, r, c, e)/Fty_d                  # yield at 1.25 x
limit
    u_u = sigma_secant(P_u, A, r, c, e)/Ftu_d                  # ultimate at 1.4 x
limit
    Pc, S = P_crit(A, r)
    u_b = (1 + MS_buck_min)*P_u/Pc                             # buckling margin >=
10 %
    u_l = 3*Fty_d/sigma_local(D, t)                            # classical local
buckling stress >= 3 x yield
    ok = (u_y <= 1) & (u_u <= 1) & (u_b <= 1) & (u_l <= 1) & (D/t <= D_over_t_max) &
(t >= t_min)
    return ok, A, dict(yield_=u_y, ult=u_u, buck=u_b, local=u_l)

def optimum(e=e_ecc, D_max=60.0, **kw):
    ok, A, u = evaluate(Dg, tg, e, **kw); ok &= (Dg <= D_max)
    m = np.where(ok, A, np.inf); i = np.unravel_index(np.argmin(m), m.shape)
    ut = {k: float(v[i]) for k, v in u.items()}
    return Dg[i], tg[i], A[i], rho*A[i]*L, ut, max(ut, key=ut.get)

if __name__ == "__main__":
    print(f"lambda={lam:.4f}  P_full={P_full/1e3:.1f} kN  P_L={P_L/1e3:.2f}
P_y={P_y/1e3:.2f}  P_u={P_u/1e3:.2f} kN")
    print("\n e(mm) Dmax(mm) | OD(mm) wall(mm)  A(mm2) mass(kg) | util yield ult buck
local | governing   [D/t<=25, t>=2.0]")
    rows = {}
    for e in (0.5, 1.0, 2.0, 5.0):
        for Dm in (40, 50, 60):
            D, t, A, m, ut, gov = optimum(e, Dm)
            print(f" {e:4.1f}   {Dm:4d}    | {D:5.1f}  {t:5.2f}   {A:6.1f}  {m:6.3f}
| {ut['yield_']:.2f} {ut['ult']:.2f} {ut['buck']:.2f} {ut['local']:.2f} | {gov}")
            rows[f"e{e}_D{Dm}"] = dict(D=float(D), t=float(t), A=float(A),
mass=float(m), gov=gov)
    json.dump(rows, open("sweep_tube_results.json", "w"), indent=1)
    print("\nSensitivity to the wall-slenderness bound (e = 1 mm, OD <= 50 mm):")
```

```
    for dt in (15, 20, 25, 30):
        D, t, A, m, ut, gov = optimum(1.0, 50, D_over_t_max=dt)
        print(f" D/t<={dt}: OD={D:.1f} t={t:.2f} A={A:.0f} mm2 mass={m:.3f} kg
util(y,u,b)=({ut['yield_']:.2f},{ut['ult']:.2f},{ut['buck']:.2f})  governing: {gov}")
    for Dsol in np.arange(15, 40, 0.5):                       # lightest solid round
bar, for comparison
        A = np.pi/4*Dsol**2; r = Dsol/4; c = Dsol/2; Pc, _ = P_crit(A, r)
        if sigma_secant(P_y, A, r, c)/Fty_d <= 1 and sigma_secant(P_u, A, r, c)/Ftu_d
<= 1 and (1 + MS_buck_min)*P_u/Pc <= 1:
            print(f"lightest solid round bar: D={Dsol} mm, A={A:.0f} mm2,
mass={rho*A*L:.3f} kg"); break
```

Appendix C. design_and_prediction.py

```
"""Baseline selection, end-fitting sizing, mass/stiffness, and test predictions for
the tubular strut."""
import numpy as np, json
from model import *
from sweep_tube import optimum, evaluate, Dg, tg      # sweep helpers

print("\n=== SENSITIVITY TO WALL-SLENDERNESS BOUND (e=1 mm, OD<=50) ===")
for dt in (15, 20, 25, 30):
    D, t, A, m, ut, gov = optimum(1.0, 50, D_over_t_max=dt)
    print(f" D/t<={dt}: OD={D:.1f} t={t:.2f} A={A:.0f} mass={m:.3f} kg
util(y,u,b)=({ut['yield_']:.2f},{ut['ult']:.2f},{ut['buck']:.2f}) gov={gov}")

# ---------------- baseline shank (D/t <= 20, OD <= 50) ----------------
D, t, A_opt, m_opt, ut, gov = optimum(1.0, 50, D_over_t_max=20)
D, t = 44.0, 2.5                                      # sweep optimum OD 44.0; wall
2.25 rounded up to 2.5 mm for boring tolerance
A, I, r, c = tube(D, t); d = D - 2*t
print(f"\nBASELINE SHANK: OD={D} wall={t} ID={d}  A={A:.1f} mm2  I={I:.0f} mm4
r={r:.3f} mm  S={K*L/r:.2f}  Cc(design)={Cc(Fty_d):.2f}")
Pc_d, S = P_crit(A, r); Pc_t, _ = P_crit(A, r, Fty_t)
print(f" Johnson Pcr design={Pc_d/1e3:.1f} kN (MS vs 1.1Pu {Pc_d/(1.1*P_u)-1:+.2f});
typical ceiling={Pc_t/1e3:.1f} kN; Euler ideal={np.pi**2*E*I/(K*L)**2/1e3:.1f} kN")
for nm, P, F in (("limit", P_L, None), ("1.25 x limit", P_y, Fty_d), ("1.4 x limit",
P_u, Ftu_d)):
    f = float(phi(P, A, r)); sg = float(sigma_secant(P, A, r, c)); dl =
float(delta_secant(P, A, r))
    print(f" {nm:13s} P={P/1e3:6.2f} kN  P/A={P/A:6.1f}  phi={f:.4f}
sec={1/np.cos(f):.4f} delta={dl:.3f} mm  sigma_max={sg:.1f} MPa" + (f"  MS={F/sg-
1:+.2f}" if F else ""))
print(f" local buckling: classical sigma_cl={sigma_local(D,t):.0f} MPa =
{sigma_local(D,t)/Fty_d:.1f} x design yield; D/t={D/t:.0f}")
print(f" shank mass per mm of tube = {rho*A:.5f} kg/mm")

# ---------------- end fittings ----------------
P_end, P_pin_el = 1.25*Pc_t, 1.10*Pc_t
print(f"\nEND REQUIREMENTS: strut ceiling {Pc_t/1e3:.1f} kN -> ends >=
{P_end/1e3:.1f} kN, pin elastic >= {P_pin_el/1e3:.1f} kN")
t_fork, g = 10.0, 0.5
best = None
for Dh in (50.0, 56.0, 60.0, 63.5):                       # head bar stock diameter
(tongue is milled from it)
```

```
    for Dp in (16, 18, 20, 22, 25, 28, 30):
        for tl in np.arange(10, 0.8*Dh+0.1, 1.0):
            w = np.sqrt(Dh**2 - tl**2); e = float(np.ceil(1.6*Dp/5)*5)
            pn = pin_capacity(Dp, t_fork, tl, g); tc = tongue_capacity(Dp, e, w, tl)
            ok = (pn['elastic'] >= P_pin_el) and (pn['ult_bend'] >= P_end) and
(pn['shear'] >= P_end) and (tc['bearing'] >= P_end) and (tc['net'] >= P_end)
            if ok:
                vol = tl*w*(e+20)
                if best is None or vol < best[0]: best = (vol, Dh, Dp, tl, w, e, pn,
tc)
vol, Dh, Dp, tl, w, e_h, pn, tc = best
print(f" minimum-volume tongue: head stock D={Dh}, pin D={Dp} mm, tongue
thickness={tl:.0f}, chord width={w:.1f}, e={e_h:.0f} (e/D={e_h/Dp:.2f},
a={e_h-Dp/2:.1f})")
Dp, tl, e_h = float(Dp), float(tl), float(e_h); w = np.sqrt(Dh**2 - tl**2)
pn = pin_capacity(Dp, t_fork, tl, g); tcd = tongue_capacity(Dp, e_h, w, tl); tct =
tongue_capacity(Dp, e_h, w, tl, Ftu=Ftu_t, Fty=Fty_t)
print(f" pin: arm={pn['arm']:.2f} mm  ult bending={pn['ult_bend']/1e3:.0f}
elastic={pn['elastic']/1e3:.0f}  shear={pn['shear']/1e3:.0f} kN")
print(f" tongue (design min): bearing={tcd['bearing']/1e3:.1f}
net={tcd['net']/1e3:.1f} kN ; typical: bearing={tct['bearing']/1e3:.1f}
net={tct['net']/1e3:.1f} kN")
for nm, cap, req in (("tongue bearing", tcd['bearing'], P_end), ("tongue net
section", tcd['net'], P_end), ("pin shear", pn['shear'], P_end), ("pin ult bending",
pn['ult_bend'], P_end), ("pin elastic", pn['elastic'], P_pin_el)):
    print(f"   {nm:19s} MS vs requirement = {cap/req-1:+.2f}   | vs ultimate design
load {cap/P_u-1:+.2f}")
print(f" tear-out sanity (2 planes, Fsu={Fsu}):
{2*Fsu*tl*(e_h-Dp/2*np.cos(np.radians(40)))/1e3:.0f} kN")

# ---------------- geometry, mass, stiffness ----------------
L_tot = L + 2*e_h; L_tongue = e_h + 20.0; L_taper = 30.0; L_tube = L_tot -
2*(L_tongue + L_taper)
R = Dh/2; zt = tl/2
A_tongue = 2*(zt*np.sqrt(R**2 - zt**2) + R**2*np.arcsin(zt/R)); A_head =
np.pi*Dh**2/4; A_tube_end = np.pi*D**2/4
vol_tube = A*L_tube
vol_taper = 2*(np.pi*L_taper/12)*(Dh**2 + Dh*D + D**2)             # frustum, solid
vol_tongue = 2*A_tongue*L_tongue - 2*np.pi/4*Dp**2*tl
mass = rho*(vol_tube + vol_taper + vol_tongue)
print(f"\nGEOMETRY: total length {L_tot:.0f}, tongue zone {L_tongue:.0f}, taper
{L_taper:.0f}, tube {L_tube:.0f} mm; head stock {Dh} mm; tube OD {D} wall {t}")
print(f"MASS = {mass:.3f} kg  (tube {rho*vol_tube:.3f}, tapers {rho*vol_taper:.3f},
tongues {rho*vol_tongue:.3f})")
A_net_h = tl*(w - Dp)
comp = L_tube/(E*A) + 2*L_taper/(E*(A_head+A_tube_end)/2) + 2*20/(E*A_net_h) +
2*(L_tongue-20)/(E*A_tongue)
print(f"AXIAL STIFFNESS: EA/L(tube, L=400)={E*A/L/1e3:.1f} kN/mm ; segmented-bar
model={1/comp/1e3:.1f} kN/mm")

# ---------------- predictions (typical properties) ----------------
print("\nPREDICTION (typical): ceiling (Johnson) = %.1f kN" % (Pc_t/1e3))
pred = {}
for ee in (0.5, 1.0, 2.0):
    fy = first_yield(A, r, c, ee); pred[ee] = fy
```

```
    print(f"  e={ee}: first yield = {fy/1e3:.1f} kN  ({fy/P_L:.2f} x limit)   delta
at limit = {float(delta_secant(P_L,A,r,ee)):.3f} mm")
print(f"  planning load for machine = 1.5 x ceiling = {1.5*Pc_t/1e3:.0f} kN =
{1.5*Pc_t/4448.2:.0f} kip ; fork bearing at that load = {1.5*Pc_t/(Dp*2*t_fork):.0f}
MPa")
print(f"  peak/mass at e=1: {pred[1.0]/1e3/mass:.0f} kN/kg")
json.dump(dict(D=D,t=t,Dh=Dh,Dp=Dp,tl=tl,e_h=e_h,mass=float(mass),ceiling=float(Pc_t)
,first_yield={str(k):float(v) for k,v in pred.items()}), open("design.json","w"),
indent=1)
```

Appendix D. figures.py

```
"""Figures: load-path / free-body diagrams, geometry drawing, sizing map, predicted
load-deflection curves."""
import numpy as np, json
import matplotlib; matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.patches import Rectangle, FancyArrowPatch, Circle, Polygon
from model import *
import sweep_tube as sw
dsg = json.load(open("design.json")); D, t, Dh, Dp, tl, e_h = dsg["D"], dsg["t"],
dsg["Dh"], dsg["Dp"], dsg["tl"], dsg["e_h"]
plt.rcParams.update({"font.size": 9})
def arr(ax, p, q, c="red", lw=2, **k): ax.add_patch(FancyArrowPatch(p, q,
arrowstyle="-|>", mutation_scale=14, color=c, lw=lw, **k))

# ---- Fig 1: free-body diagrams ----
fig, ax = plt.subplots(1, 3, figsize=(13, 4.6))
a = ax[0]; a.set_xlim(0, 10); a.set_ylim(0, 8); a.axis("off"); a.set_title("(a)
Vehicle-level load path")
a.add_patch(Rectangle((1, 6.2), 8, 0.6, fc="#dddddd", ec="k")); a.text(5, 6.9,
"thrust frame / tank bulkhead", ha="center")
a.add_patch(Polygon([[4.3, 2.0], [5.7, 2.0], [6.1, 0.6], [3.9, 0.6]], fc="#bbbbbb",
ec="k")); a.text(5, 0.2, "engine", ha="center")
for x0 in (1.8, 3.4, 6.6, 8.2):
    a.plot([x0, 4.6 if x0 < 5 else 5.4], [6.2, 2.0], "k-", lw=3)
arr(a, (5, 1.2), (5, 5.2), "tab:blue", 3); a.text(5.2, 4.0, "T", color="tab:blue",
fontsize=12)
a.text(0.2, 1.2, "N = 4 struts\nat angle alpha = 20 deg\nP = T / (N cos alpha)",
fontsize=9)
b = ax[1]; b.set_xlim(0, 10); b.set_ylim(0, 8); b.axis("off"); b.set_title("(b)
Strut: two-force member, eccentric load")
b.plot([5, 5], [1, 7], "k-", lw=7)
xs = np.linspace(1, 7, 50); b.plot(5 + 0.5*np.sin(np.pi*(xs-1)/6), xs, "b:", lw=2)
arr(b, (5.8, 7.6), (5.8, 7.05), "red"); arr(b, (5.8, 0.4), (5.8, 0.95), "red")
b.text(6.2, 7.5, "P", color="red", fontsize=12); b.text(6.2, 0.2, "P", color="red",
fontsize=12)
b.annotate("", xy=(5, 7.3), xytext=(5.8, 7.3), arrowprops=dict(arrowstyle="<->"));
b.text(5.3, 7.75, "e", fontsize=11)
b.annotate("", xy=(5, 4), xytext=(5.5, 4), arrowprops=dict(arrowstyle="<->",
color="b")); b.text(5.6, 4, "delta", color="b")
b.annotate("", xy=(3.2, 1), xytext=(3.2, 7), arrowprops=dict(arrowstyle="<->"));
b.text(0.2, 4, "L = 400 mm\n(pin to pin)", fontsize=9)
b.text(8.0, 5.5, "sigma_max = P/A [1 + (e c/r^2)\n sec(KL/2r sqrt(P/AE))]",
fontsize=8, ha="center")
```

```
c = ax[2]; c.set_xlim(0, 12); c.set_ylim(0, 8); c.axis("off"); c.set_title("(c) Pin
joint: double shear")
c.add_patch(Rectangle((1, 3.2), 4.5, 1.6, fc="#cfe2f3", ec="k")); c.text(2.8, 4.0,
"strut tongue\n(thickness t_l)", ha="center", va="center")
for y0 in (4.9, 2.3): c.add_patch(Rectangle((4, y0), 5, 0.9, fc="#eeeeee", ec="k"))
c.text(7.8, 5.35, "fork plate (t1)", ha="center"); c.text(7.8, 2.75, "fork plate
(t1)", ha="center")
c.add_patch(Rectangle((4.6, 2.3), 0.6, 3.5, fc="#999999", ec="k")); c.text(4.9, 1.8,
"pin", ha="center")
arr(c, (0.2, 4.0), (1.0, 4.0), "red"); c.text(0.1, 4.4, "P", color="red",
fontsize=12)
arr(c, (9.8, 5.35), (9.1, 5.35), "red"); arr(c, (9.8, 2.75), (9.1, 2.75), "red");
c.text(9.9, 5.6, "P/2", color="red"); c.text(9.9, 3.0, "P/2", color="red")
c.text(6.0, 0.7, "pin bending arm = t1/2 + t_l/4 + g", ha="center")
plt.tight_layout(); plt.savefig("fig1_free_body_diagrams.png", dpi=170); plt.close()

# ---- Fig 2: geometry ----
L_tot = L + 2*e_h; Lt = e_h + 20.0; Lk = 30.0
fig, ax = plt.subplots(figsize=(13, 3.6)); ax.set_aspect("equal"); ax.axis("off")
xs_ = [0, Lt, Lt+Lk, L_tot-Lt-Lk, L_tot-Lt, L_tot]
ax.add_patch(Polygon([[xs_[0], -tl/2], [xs_[1], -tl/2], [xs_[1], -Dh/2], [xs_[2], -
D/2], [xs_[3], -D/2], [xs_[4], -Dh/2], [xs_[4], -tl/2], [xs_[5], -tl/2], [xs_[5],
tl/2], [xs_[4], tl/2], [xs_[4], Dh/2], [xs_[3], D/2], [xs_[2], D/2], [xs_[1], Dh/2],
[xs_[1], tl/2], [xs_[0], tl/2]], fc="#cfe2f3", ec="k"))
ax.add_patch(Rectangle((xs_[2]+0.0, -(D/2-t)), xs_[3]-xs_[2], D-2*t, fc="white",
ec="k", ls="--"))
for xc in (e_h, L_tot - e_h):
    ax.add_patch(Circle((xc, 0), 2.0, fc="k"))
    ax.plot([xc, xc], [-Dh/2-4, Dh/2+4], "k-.", lw=0.6)
ax.annotate("", xy=(e_h, -Dh/2-8), xytext=(L_tot-e_h, -Dh/2-8),
arrowprops=dict(arrowstyle="<->")); ax.text(L_tot/2, -Dh/2-14, f"pin centre distance
L = {L:.0f} mm", ha="center")
ax.annotate("", xy=(0, Dh/2+8), xytext=(L_tot, Dh/2+8),
arrowprops=dict(arrowstyle="<->")); ax.text(L_tot/2, Dh/2+12, f"overall length
{L_tot:.0f} mm", ha="center")
ax.text(L_tot/2, 0, f"tube OD {D:.0f} x wall {t:.1f} (bored)", ha="center",
va="center", fontsize=9)
ax.text(Lt/2+5, Dh/2+0, f"tongue {tl:.0f} mm thick\npin hole {Dp:.0f} H7, e =
{e_h:.0f} mm", ha="center", va="top", fontsize=8)
ax.set_xlim(-20, L_tot+20); ax.set_ylim(-Dh/2-22, Dh/2+22); ax.set_title("Strut
geometry, edge view of the tongues (schematic; shoulder radii not drawn)")
plt.tight_layout(); plt.savefig("fig2_geometry.png", dpi=170); plt.close()

# ---- Fig 3: sizing map and constraint utilisation ----
ok, A_, u = sw.evaluate(sw.Dg, sw.tg, 1.0, D_over_t_max=20)
fig, ax = plt.subplots(1, 2, figsize=(12, 4.4))
ax[0].contourf(sw.Dg, sw.tg, ok, levels=[-.5, .5, 1.5], colors=["#f4cccc",
"#d9ead3"])
cs = ax[0].contour(sw.Dg, sw.tg, rho*A_*L, levels=[0.25, 0.3, 0.35, 0.4, 0.5, 0.6],
colors="gray", linewidths=.7); ax[0].clabel(cs, fmt="%.2f kg")
ax[0].contour(sw.Dg, sw.tg, u["buck"], levels=[1], colors="b", linestyles="--");
ax[0].contour(sw.Dg, sw.tg, np.maximum(u["yield_"], u["ult"]), levels=[1],
colors="r")
ax[0].plot(D, t, "ko"); ax[0].plot(44, 2.25, "k^")
ax[0].set_xlabel("tube outside diameter (mm)"); ax[0].set_ylabel("wall thickness
```

```
(mm)"); ax[0].set_ylim(1.5, 8); ax[0].set_xlim(28, 60)
ax[0].set_title("Feasible region (green, e = 1 mm, D/t <= 20)\nblue dashed: buckling
limit, red: strength limit")
Dv = np.arange(30, 60.1, 0.5); tv = np.full_like(Dv, 2.5)
okv, Av, uv = sw.evaluate(Dv, tv, 1.0, D_over_t_max=100)
ax[1].plot(Dv, uv["yield_"], label="yield (1.25 x limit)"); ax[1].plot(Dv, uv["ult"],
label="ultimate (1.4 x limit)"); ax[1].plot(Dv, uv["buck"], label="buckling (MS >=
10%)")
ax[1].axhline(1, color="k", lw=.8); ax[1].set_ylim(0, 2); ax[1].set_xlabel("tube OD
at 2.5 mm wall (mm)"); ax[1].set_ylabel("demand / allowable"); ax[1].legend();
ax[1].set_title("Active criterion along the 2.5 mm wall line")
plt.tight_layout(); plt.savefig("fig3_sizing_map.png", dpi=170); plt.close()

# ---- Fig 4: predicted load vs lateral offset ----
A, I, r, c = tube(D, t); fig, ax = plt.subplots(figsize=(6.4, 4.4))
Pc_t, _ = P_crit(A, r, Fty_t)
for e in (0.5, 1.0, 2.0):
    Ps = np.linspace(0, 0.995*Pc_t, 800); ax.plot(delta_secant(Ps, A, r, e) + e,
Ps/1e3, label=f"e = {e} mm")
ax.axhline(P_L/1e3, color="g", ls=":", label="limit load"); ax.axhline(P_u/1e3,
color="orange", ls="--", label="1.4 x limit")
ax.axhline(Pc_t/1e3, color="k", lw=.7, label="Johnson ceiling (typical)")
ax.set_xlim(0, 8); ax.set_xlabel("mid-length lateral offset (mm)");
ax.set_ylabel("axial load (kN)"); ax.set_title("Predicted load vs lateral offset
(secant model)"); ax.legend(fontsize=8)
plt.tight_layout(); plt.savefig("fig4_prediction.png", dpi=170); plt.close()
print("figures written")
```

Appendix E. validation_targets.py

This script imports model.py (Appendix A). It produces the hand-calculation targets of Section 9.6, compares the eigenvalue results with the hand values of Section 9.7, and gives the modal targets of Section 9.8. It is delivered as a file with the same name.

```
"""Hand-calculation targets for the FEA validation (Section 9) and test predictions.
Uses model.py.  Units N, mm, MPa."""
import numpy as np
from model import *
A, I, r, c = tube(44, 2.5)
print(f"tube A={A:.1f} mm2  I={I:.0f} mm4  r={r:.3f} mm  c={c} mm  S={K*L/r:.2f}")
print(f"axial stress at limit P/A = {P_L/A:.1f} MPa ; axial strain =
{P_L/(E*A):.3e} ; shortening of 150 mm tube half = {P_L*150/(E*A):.3f} mm")
print("\n--- eccentricity cases at the limit load (42.5 kN) ---")
print(" e(mm) | end moment P*e (N.m) | phi | sec | added deflection (mm) | mid moment
P(e+d) (N.m) | sigma concave | sigma convex (MPa)")
for e in (0.0, 0.5, 1.0, 2.0):
    f = float(phi(P_L, A, r)); sec = 1/np.cos(f); d = e*(sec-1); Mm = P_L*(e+d)
    sc = P_L/A*(1 + e*c/r**2*sec); sv = P_L/A*(1 - e*c/r**2*sec)
    print(f" {e:4.1f}  | {P_L*e/1e3:8.1f}            | {f:.4f} | {sec:.4f} | {d:7.3f}
| {Mm/1e3:8.1f}               | {sc:7.1f}   | {sv:7.1f}")
print("\n--- same at 1.25 x and 1.4 x limit (e = 2 mm) ---")
for nm, P in (("1.25x", P_y), ("1.4x", P_u)):
    f = float(phi(P, A, r)); sec = 1/np.cos(f); print(f" {nm}: P={P/1e3:.1f} kN
sigma concave={P/A*(1+2*c/r**2*sec):.1f} MPa  delta={2*(sec-1):.3f} mm
```

```
M_end={P*2/1e3:.1f} N.m")
print("\n--- first yield and ceiling (typical Fty = 276 MPa) ---")
print(f" ceiling (Johnson) = {P_crit(A,r,Fty_t)[0]/1e3:.1f} kN")
for e in (0.5, 1.0, 2.0): print(f" e={e}: first yield {first_yield(A,r,c,e)/1e3:.1f}
kN")
print("\n--- buckling ---")
PE = np.pi**2*E*I/(K*L)**2
print(f" Euler (K=1) = {PE/1e3:.1f} kN -> load multiplier at 42.5 kN = {PE/P_L:.2f}")
print(f" classical local buckling stress = {sigma_local(44,2.5):.0f} MPa -> load
multiplier = {sigma_local(44,2.5)/(P_L/A):.1f}")
print("\n--- comparison with the Ansys eigenvalue results (e = 2 mm pre-stress) ---")
for nm, lm in (("mode 1", 5.9653), ("mode 2", 16.627), ("mode 3/4 (approx)", 24.6),
("mode 10 (local)", 29.812)):
    print(f" {nm}: LM={lm:.3f} -> critical load {lm*P_L/1e3:.1f} kN ; ratio to mode 1
= {lm/5.9653:.2f}")
print(f" K_eff from mode 1 = {np.sqrt(PE/(5.9653*P_L)):.3f} ; implied K of mode 2 =
{np.sqrt(PE/(16.627*P_L)):.3f}")
print(f" FEA/Euler (mode 1) = {5.9653*P_L/PE:.3f} ; local mode: stress equivalent
{29.812*P_L/A:.0f} MPa = {29.812*P_L/A/sigma_local(44,2.5):.2f} x classical")
print("\n--- modal targets (uniform tube approximation, E=68.9 GPa, rho=2700) ---")
E_, I_, A_, rho_ = 68.9e9, I*1e-12, A*1e-6, 2700.0
f0 = np.pi/(2*0.4**2)*np.sqrt(E_*I_/(rho_*A_)); print(f" pin-pin lateral f1 (L=0.4 m)
= {f0:.0f} Hz ; with axial load: " + ", ".join(f"{P/1e3:.1f} kN:
{f0*np.sqrt(1-P/PE):.0f} Hz" for P in (P_L, P_u, 83.2e3)))
Lm = 0.47; print(f" free-free bending f1 (uniform tube, L={Lm} m) =
{4.730**2/(2*np.pi*Lm**2)*np.sqrt(E_*I_/(rho_*A_)):.0f} Hz ; axial free-free f1 =
{np.sqrt(E_/rho_)/(2*Lm):.0f} Hz")9
```